

AI PERSONAL CFO

Software Requirements Specification & Master Engineering Blueprint

Android-first, AI-driven personal finance application for India

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1. Executive Summary & Vision

AI Personal CFO is an Android-first, AI-driven personal finance application designed to act as a full-time Chief Financial Officer for an individual or household in India. Where conventional expense trackers merely record what already happened, AI Personal CFO is built to **understand, predict, and advise**: it classifies spending automatically, forecasts cash flow, warns about upcoming financial stress, recommends emergency-fund and investment actions, evaluates major purchases before the user commits, and explains every recommendation in plain language.

The product is **privacy-first and offline-first**. All financial data lives in an encrypted local database on the device; cloud services are optional, opt-in, and end-to-end encrypted. AI runs as a layered pipeline — deterministic rules and statistical models produce the numbers, and a large language model (LLM) is used only to explain and converse, never to invent figures.

1.1 Vision Statement

"Every person deserves a CFO. AI Personal CFO gives each user a tireless, transparent, and trustworthy financial brain that knows their complete financial picture, anticipates problems before they occur, and turns raw transactions into confident decisions."

1.2 Problem Statement

- Most Indians manage money reactively: salary arrives, spending happens, and the balance at month-end is a surprise.
- Existing apps are ledgers, not advisors — they show charts but do not answer "Can I afford this bike?" or "Will I run out of money before the 28th?"
- Financial data is fragmented across bank SMS, UPI apps, credit-card statements, paper receipts, gold, vehicles, and informal cash.
- Personalised financial advice is expensive, generic, or conflicted (commission-driven). Users need advice that is computed from their own data with transparent reasoning.
- Users distrust cloud finance apps with raw account credentials; a local-first model removes that objection.

1.3 Product Principles (binding for all design and code decisions)

RULE / NOTE: These principles override feature requests. Any feature that violates a principle must be redesigned or rejected during review.

ID	Principle	Practical Rule
P-01	Privacy first	No financial data leaves the device without explicit, revocable, per-feature consent. Default = fully offline.
P-02	AI must show its work	Every recommendation displays the inputs, the rule/model that fired, and a plain-language explanation. No black-box verdicts.
P-03	Numbers come from math, words come from AI	LLMs never compute amounts, scores, or forecasts. They only verbalise results produced by deterministic engines.

ID	Principle	Practical Rule
P-04	Offline-first	Every core feature works with airplane mode on. Network features degrade gracefully with cached data and staleness labels.
P-05	Zero data entry where possible	OCR, SMS parsing (opt-in), recurring detection, and smart defaults minimise typing.
P-06	India-native	INR and lakh/crore formatting, UPI-centric flows, RBI rates, Indian gold/fuel prices, Indian tax and instrument awareness (FD, RD, PPF, EPF, NPS, ELSS).
P-07	Advice, never orders	The app recommends and simulates; the user decides. No auto-executed financial actions in v1.
P-08	Deterministic and testable	Every engine must be unit-testable with fixed inputs producing fixed outputs. Randomness only via injected, seedable sources.

1.4 What the Product Is / Is Not

In scope (the product IS)	Out of scope for v1 (the product IS NOT)
A personal/household finance brain: tracking, budgeting, forecasting, advising	A broker, bank, wallet, or payment app — it never moves money
Manual + OCR + (opt-in) SMS-parsed transaction capture	Automatic bank account aggregation via credentials/screen-scraping
Rule + statistics + on-device ML predictions with LLM explanations	A SEBI-registered investment advisor giving regulated advice
Local encrypted storage with optional encrypted backup	A cloud service that stores plaintext financial data
Single user + optional household profiles on one device	Multi-tenant social features, leaderboards, ads, data resale

2. Business Goals & Success Metrics

2.1 Business Goals

ID	Goal	Measure
BG-01	Become the user's single source of financial truth	≥ 80% of active users log/import ≥ 90% of monthly spends by month 3 of use
BG-02	Deliver advice users act on	≥ 40% of AI recommendations marked "followed" or "helpful"
BG-03	Habit formation	Day-30 retention ≥ 35%; median ≥ 4 sessions/week
BG-04	Trust through transparency	≥ 4.5 average rating on privacy/trust survey questions; < 1% data-related complaints
BG-05	Sustainable monetisation (post v1)	Freemium: free core ledger; Pro subscription unlocks advanced AI (forecast horizon, simulations, household mode)
BG-06	Low support cost	Self-serve resolution ≥ 90% via in-app help + AI assistant

2.2 Product KPIs (instrumented, privacy-safe)

- **Activation:** user reaches "aha" = first AI insight viewed within 10 minutes of install.
- **Coverage:** % of user's real spending captured (self-reported calibration prompt monthly).
- **Forecast accuracy:** MAPE of 30-day cash-flow forecast vs actuals, computed on-device; target ≤ 15% after 3 months of data.
- **Classification accuracy:** % of auto-categorised transactions not re-edited by user; target ≥ 92%.
- **Advice engagement:** recommendation open rate, follow rate, dismiss-with-reason rate.
- All KPIs are computed on-device; only anonymised, aggregated counters are uploaded, and only if analytics consent is ON (default OFF).

2.3 Non-Goals for v1

- iOS app (planned Phase 5; architecture must keep domain layer in portable Kotlin for future KMP reuse).
- Account Aggregator (AA) framework integration — designed for, but not built in v1 (see §27 and Roadmap).
- Automated tax filing, insurance sales, or loan origination.

3. Target Users & Personas

Primary market: urban and semi-urban Indian smartphone users, age 22–45, salaried or self-employed, digitally comfortable (UPI users), financially aspirational but not financially expert.

3.1 Persona 1 — "Arjun", the First-Salary Professional (Primary)

Attribute	Detail
Profile	24, software engineer in Bengaluru, ₹85k/month, single, rents a flat, has a two-wheeler
Behaviour	Pays for everything by UPI; money "disappears"; has 1 credit card he fears; saves randomly
Goals	Build an emergency fund, start SIPs, buy a better bike in 18 months
Pain points	No visibility of where money goes; doesn't know how much is safe to spend today
Key features	Auto-categorisation, Safe-to-Spend widget, Emergency Fund coach, Purchase Advisor, cash-flow forecast
Success moment	"The app told me I could afford the bike in March, not now — and showed me exactly why."

3.2 Persona 2 — "Meera & Ravi", the Family CFO (Primary)

Attribute	Detail
Profile	34 & 36, dual income ₹2.1L/month, one child, home loan, car, gold, MF portfolio, parents to support
Behaviour	Meera manages daily budget, Ravi manages investments; they reconcile monthly in a spreadsheet
Goals	Child's education corpus, prepay home loan vs invest decision, keep EMI ratio healthy
Pain points	Fragmented picture; goal math is hard; car service costs always surprise them
Key features	Household profile, goal engine, loan vs invest simulator, vehicle maintenance prediction, health score
Success moment	"One dashboard finally shows our whole net worth, and it warned us about the insurance premium due next month."

3.3 Persona 3 — "Suresh", the Variable-Income Freelancer (Secondary)

Attribute	Detail
Profile	29, freelance designer, income ₹40k–₹1.6L/month, irregular; GST-registered
Behaviour	Feast-and-famine spending; keeps buffer in savings account "just in case"
Goals	Smooth income, know minimum monthly needs, separate business vs personal
Pain points	Fixed advice ("save 20% of salary") doesn't fit irregular income
Key features	Fixed-vs-variable expense split, income volatility model, larger emergency-fund multiplier, tagging (business/personal)

3.4 Persona 4 — "Lakshmi", the Cash-Heavy Saver (Secondary)

Attribute	Detail
Profile	41, runs a small boutique, mixes cash and UPI, buys gold regularly, low trust in cloud apps
Key features	Fast manual/cash entry, receipt OCR, gold asset tracking with live prices, fully offline mode, local backup to her own storage

3.5 Accessibility & Inclusivity Requirements

- Full TalkBack support; every interactive element has a content description (enforced by lint + UI tests).
- Dynamic type up to 200% without layout breakage; minimum touch target 48dp.
- Colour is never the only signal (icons/labels accompany red/green states); WCAG 2.1 AA contrast.
- Language: English at launch; all strings externalised and ICU-plural-ready for Hindi/Kannada/Tamil in Phase 4.
- Number formatting follows Indian system (1,00,000 = 1 lakh) with a settings toggle for international format.

4. Document Conventions & Requirement Rules

4.1 Requirement Identification

- **FR-<area>-<nnn>** — functional requirement (e.g., FR-TXN-012).
- **NFR-<nnn>** — non-functional requirement.
- **AI-<engine>-<nnn>** — AI/algorithm requirement.
- **DB-<nnn>** — data model requirement. **SEC-<nnn>** — security requirement.
- Every requirement uses RFC-2119 keywords: **MUST**, **SHOULD**, **MAY**. **MUST** items block release.

4.2 Definition of Done (applies to every feature)

- Requirement implemented and traceable (commit message references FR id).
- Unit tests for domain logic ($\geq 85\%$ coverage on engine modules; 100% on money math).
- UI state covered by at least one Compose UI test or screenshot test.
- Works offline; verified with airplane-mode test case.
- Accessibility scan passes; strings externalised; dark mode verified.
- No new lint/detekt warnings; CI green; reviewed by at least one other developer (or self-review checklist if solo).

4.3 Glossary (selected)

Term	Meaning
Safe-to-Spend (STS)	Money available today after reserving upcoming fixed obligations, goal contributions, and buffer
Fixed expense	Recurring, low-variance, obligation-like outflow (rent, EMI, subscriptions)
Variable expense	Discretionary or fluctuating outflow (food, shopping, travel)
FHS	Financial Health Score (0–1000), §14
Engine	A pure-Kotlin, deterministic domain module producing typed results (no Android dependencies)
Insight	A generated, user-visible card with a finding, evidence, and optional action
Snapshot	Immutable daily record of balances/net worth used for history and forecasting

5. Functional Requirements — Screen by Screen

The app is organised into five bottom-navigation destinations — **Home**, **Money (transactions)**, **Plan (budgets & goals)**, **Wealth (accounts & investments)**, **Advisor (AI)** — plus modal flows (add transaction, scan receipt, onboarding, settings). Each screen below lists purpose, key UI elements, and binding requirements.

5.1 Onboarding & Setup Flow

ID	Requirement	Priority
FR-ONB-001	App MUST be usable after a 4-step onboarding: (1) welcome & privacy pledge, (2) create local profile + choose currency format, (3) set up security (biometric/PIN), (4) add first account with opening balance.	MUST

ID	Requirement	Priority
FR-ONB-002	Onboarding MUST offer optional quick-setup: monthly income, rent/EMI amount, typical savings — used to seed budgets and the emergency-fund target.	MUST
FR-ONB-003	SMS-parsing permission MUST be presented as a separate, skippable, clearly-explained opt-in with examples of what is read and the statement that parsing is on-device only.	MUST
FR-ONB-004	A demo mode with realistic sample data MUST be available without creating a profile.	SHOULD
FR-ONB-005	Onboarding completion in under 3 minutes for the default path (measured in analytics if consented).	SHOULD

5.2 Home — Unified Dashboard

Purpose: answer three questions in five seconds — **How am I doing? What can I spend today? What needs my attention?**

ID	Requirement	Priority
FR-HOME-001	Dashboard MUST show: net worth (with 30-day sparkline), Safe-to-Spend today, month-to-date spend vs budget, upcoming obligations (next 14 days), top 3 AI insights, and goal progress chips.	MUST
FR-HOME-002	Every dashboard number MUST be tappable and navigate to its detail screen.	MUST
FR-HOME-003	A privacy toggle (eye icon) MUST blur all amounts instantly; state persists per session.	MUST
FR-HOME-004	Dashboard MUST render fully from local data in < 500 ms on a mid-range device (cold start < 2 s to first frame).	MUST
FR-HOME-005	Cards MUST be reorderable/hideable via edit mode; layout stored per profile.	SHOULD
FR-HOME-006	A home-screen Android widget MUST show Safe-to-Spend and month spend without opening the app (respects privacy blur setting).	SHOULD
FR-HOME-007	Pull-to-refresh recomputes insights and refreshes external market data if online; stale external data MUST show "as of <date>".	MUST

5.3 Money — Transactions List & Detail

ID	Requirement	Priority
FR-TXN-001	Transactions MUST support: amount, currency, date-time, account, category, subcategory, payee/merchant, notes, tags (many), attachments (many), type (expense/income/transfer), and split lines.	MUST
FR-TXN-002	Add-transaction MUST be reachable in one tap (FAB) and completable in ≤ 3 taps for a common expense (amount → category suggestion → save).	MUST
FR-TXN-003	Transfers MUST be a single logical record affecting two accounts atomically; deleting one side deletes both.	MUST
FR-TXN-004	Split transactions MUST allow N lines with independent categories; lines MUST sum exactly to the parent amount (validated, no rounding drift).	MUST
FR-TXN-005	Auto-categorisation MUST run on every new transaction (see AI-CLS); user correction MUST retrain the personal model and offer "apply to similar past transactions".	MUST

ID	Requirement	Priority
FR-TXN-006	Recurring detection MUST propose recurring series when ≥ 2 similar transactions match on merchant/amount/cadence; user confirms to create a Recurring Rule.	MUST
FR-TXN-007	List MUST support search (payee, note, amount, tag), filters (date range, account, category, type, amount range), grouping by day with daily totals, and infinite scroll via paging.	MUST
FR-TXN-008	Bulk edit MUST support multi-select recategorise, retag, delete (with undo snackbar, 7-day soft delete).	SHOULD
FR-TXN-009	Every transaction MUST record its source: manual, ocr, sms, import, recurring-auto.	MUST
FR-TXN-010	Future-dated transactions MUST be supported and excluded from actuals but included in forecasts.	MUST

5.4 Scan — OCR Receipt Capture (modal flow)

ID	Requirement	Priority
FR-OCR-001	Camera capture and gallery import MUST both be supported; multi-page receipts MAY be stitched.	MUST
FR-OCR-002	OCR MUST run fully on-device (ML Kit Text Recognition v2); no receipt image ever leaves the device unless the user explicitly shares it.	MUST
FR-OCR-003	The parser MUST extract: total amount, date, merchant name, and (best-effort) tax and line items; each field shows a confidence indicator and is editable before save.	MUST
FR-OCR-004	Extraction review screen MUST prevent saving without an amount and date; low-confidence fields are visually flagged.	MUST
FR-OCR-005	The original image MUST be stored encrypted and linked as an attachment; user can delete image while keeping the transaction.	MUST
FR-OCR-006	Duplicate guard: if an SMS/manual transaction with same amount $\pm 1\%$ and date ± 1 day exists, MUST offer merge instead of creating a duplicate.	MUST

5.5 Plan — Budgets

ID	Requirement	Priority
FR-BUD-001	Budgets MUST be definable per category or category-group, per month, with optional rollover of unused amount.	MUST
FR-BUD-002	AI budget suggestions MUST be offered from 3 months of history (median + seasonality adjustment) with one-tap accept per category.	MUST
FR-BUD-003	Budget screen MUST show: spent, remaining, daily pace vs safe pace, and projected end-of-month figure per budget.	MUST
FR-BUD-004	Overspend alerts MUST fire at 80% (warning) and 100% (exceeded), and a pace alert when projected overshoot $> 10\%$ before the 20th.	MUST
FR-BUD-005	A monthly budget review flow MUST summarise wins/misses and propose next month's numbers.	SHOULD

5.6 Plan — Goals

ID	Requirement	Priority
FR-GOAL-001	Goal types MUST include: Emergency Fund (system goal, always present), Purchase, Travel, Home Down-payment, Education, Retirement, Debt Payoff, Custom.	MUST
FR-GOAL-002	Each goal MUST have: target amount, target date (or open), priority, funding accounts, expected return assumption, and computed required monthly contribution (see §15).	MUST
FR-GOAL-003	Goal feasibility MUST be checked against forecast surplus; infeasible plans show the gap and three levers (extend date, reduce target, increase contribution).	MUST
FR-GOAL-004	Contributions MUST be linkable to real transactions/accounts so progress is evidence-based, not self-declared.	MUST
FR-GOAL-005	Goal conflicts (combined contributions exceed surplus) MUST trigger a prioritisation assistant.	SHOULD

5.7 Wealth — Accounts & Net Worth

ID	Requirement	Priority
FR-ACC-001	Account types MUST include: bank (savings/current), cash, credit card, loan (home/vehicle/personal/education), investment (MF/stocks/FD/RD/PPF/EPF/NPS), gold (physical/digital/SGB), crypto, property, vehicle, receivable/payable (informal lending).	MUST
FR-ACC-002	Credit cards MUST track: limit, statement day, due day, current unbilled, last statement amount; utilisation feeds the health score.	MUST
FR-ACC-003	Loans MUST store principal, rate, EMI, tenure and expose an amortisation schedule, payoff date, and prepayment simulator.	MUST
FR-ACC-004	Asset accounts (gold, property, vehicle, crypto, MF) MUST support valuation history; market-priced assets auto-revalue from external data when online (§17).	MUST
FR-ACC-005	Net worth MUST equal assets – liabilities, snapshotted daily (DB \$10 snapshots table), with 1M/6M/1Y/All charts.	MUST
FR-ACC-006	Account reconciliation flow MUST let user enter real balance; difference is posted as an adjustment transaction (never silently mutated).	MUST
FR-ACC-007	Archived/closed accounts MUST retain history and be excluded from active totals.	MUST

5.8 Wealth — Investments & Vehicle

ID	Requirement	Priority
FR-INV-001	Investment holdings MUST record units, average cost, and current price; returns shown as absolute and XIRR.	MUST
FR-INV-002	Portfolio view MUST show allocation by asset class, diversification analysis, and concentration warnings (§12).	MUST
FR-VEH-001	Vehicle profile MUST capture: type, make/model/year, purchase price/date, odometer log, fuel log (litres, price, odo), insurance & PUC expiry, service history.	MUST
FR-VEH-002	Fuel log MUST compute mileage (km/l) trends; anomalous mileage drop triggers a maintenance insight (§13).	MUST
FR-VEH-003	Vehicle screen MUST show true monthly cost of ownership (fuel + EMI + insurance amortised + service + depreciation).	SHOULD

5.9 Advisor — AI Hub, Chat, Purchase Advisor

ID	Requirement	Priority
FR-AI-001	Advisor hub MUST list: Financial Health Score with drill-down, insight feed (dismiss / snooze / act), Purchase Advisor entry, simulators, and chat.	MUST
FR-AI-002	Every insight card MUST include: finding, evidence ("based on 14 transactions in Dining, Apr–Jun"), confidence, and at most one recommended action.	MUST
FR-AI-003	Purchase Advisor MUST accept item, price, payment method (cash/EMI/card), urgency; output a verdict (Comfortable / Stretch / Not now) with full reasoning trace (§13-PA).	MUST
FR-AI-004	Chat assistant MUST answer questions about the user's own data via the tool-calling design in §19, and MUST refuse to fabricate numbers when a tool cannot answer.	MUST
FR-AI-005	All AI outputs MUST carry the disclaimer role: informational, not regulated investment advice; shown in UI footer of Advisor screens.	MUST

5.10 Settings, Security, Data

ID	Requirement	Priority
FR-SET-001	Settings MUST include: profile & household, categories editor, recurring rules manager, notification preferences (per type + quiet hours), security (biometric, PIN, auto-lock timer), data (backup, restore, export CSV/JSON, import, erase all), consents dashboard, about/licences.	MUST
FR-SET-002	Export MUST produce a complete, documented JSON archive plus CSVs; import MUST round-trip losslessly.	MUST
FR-SET-003	"Erase all data" MUST require biometric + typed confirmation and MUST cryptographically destroy the DB key.	MUST
FR-SET-004	Consents dashboard MUST list every permission/consent with status, plain-language purpose, and one-tap revoke.	MUST

6. Non-Functional Requirements

ID	Requirement	Target
NFR-001	Cold start to interactive dashboard	< 2.0 s (P90, mid-range device)
NFR-002	Add-transaction save latency	< 100 ms local commit
NFR-003	Full insight recompute for 5 years / 50k transactions	< 3 s background, never blocks UI
NFR-004	Crash-free sessions	≥ 99.8%
NFR-005	App size	≤ 40 MB base APK (ML models via dynamic delivery if needed)
NFR-006	Battery	Background work only via WorkManager, batched, network-aware; no wakelocks
NFR-007	Data durability	Zero data loss on process death; WAL mode; all writes transactional
NFR-008	Min Android version	API 26 (Android 8.0); target latest stable API
NFR-009	Offline behaviour	100% of core flows function offline; external data cached with TTL + staleness labels
NFR-010	Accessibility	WCAG 2.1 AA equivalent; TalkBack complete
NFR-011	Localisation readiness	No hardcoded strings; ICU plurals; RTL-safe layouts
NFR-012	Money precision	All money stored as integer minor units (paise); no floating point in money math — ever

RULE / NOTE: NFR-012 is a code-review blocking rule. Any Double/Float touching a monetary value fails review. Use the Money value class (§21.4).

7. AI Architecture Overview

The AI system is a **layered pipeline**, not a single model. Each layer has a single responsibility, a typed contract, and can be tested in isolation. Determinism increases toward the bottom; language fluency increases toward the top.

L6 CONVERSATION LAYER (LLM)

	Chat assistant, explanation generation, NL → intent Never computes numbers. Consumes structured results.
L5	DECISION ENGINE Purchase Advisor, recommendations, action ranking Combines L3/L4 outputs with user policy & priorities
L4	PREDICTION MODELS Cash-flow forecast, seasonality, maintenance prediction, spend projection, income volatility
L3	RULES ENGINE Threshold alerts, budget rules, health-score rubric, eligibility checks, guardrails on all AI output
L2	ANALYTICS / FEATURE LAYER Aggregations, category stats, recurring detection, fixed-vs-variable classification, feature vectors
L1	DATA LAYER Encrypted Room DB – single source of truth

7.1 Binding Architecture Rules

ID	Rule
AI-ARC-001	Layer N may depend only on layers below it. No layer reads the UI, and the UI reads only published results (Insight, Forecast, Verdict objects).
AI-ARC-002	Every engine is a pure Kotlin module: input data classes in → result data class out. No Android framework imports; clock, locale, and randomness injected.
AI-ARC-003	Every engine result carries provenance: engineId, engineVersion, inputWindow, computedAt, confidence, and the evidence list shown to users.
AI-ARC-004	LLM output MUST pass through the Rules Engine guardrail before display: numeric claims are verified against engine results; unverifiable numbers are stripped and the response regenerated or refused.
AI-ARC-005	All engines run on background dispatchers via a single InsightOrchestrator; results are persisted so the UI never awaits computation.
AI-ARC-006	Engine versions are recorded with every stored result so historical insights remain reproducible after algorithm upgrades.
AI-ARC-007	LLM inference is on-device by default (compact instruction model, e.g. Gemma-class via MediaPipe/AICore). Cloud LLM is optional, off by default, and receives only the minimal structured context defined in §19.4 — never raw transaction dumps.

7.2 Insight Orchestrator

- Triggers: transaction committed (debounced 5 s), day rollover, external data refresh, manual pull-to-refresh, weekly deep-analysis job.
- Pipeline: build feature snapshot (L2) → run rules (L3) → run due predictions (L4) → decision ranking (L5) → persist Insights → notify (subject to notification policy §18).
- Budget: incremental recompute after a transaction < 200 ms; weekly deep job < 3 s for 50k transactions.
- Deduplication: an insight with the same fingerprint (type + subject + period) is updated, not duplicated; dismissed fingerprints are suppressed for their snooze window.

8. Fixed vs Variable Expense Classification (AI-CLS)

Purpose: label every expense stream as **Fixed**, **Variable**, or **Semi-fixed**, because every downstream engine (Safe-to-Spend, emergency fund, forecasting, purchase advisor) depends on knowing which outflows are obligations.

8.1 Two-stage design

- **Stage 1 — Category assignment** (per transaction): hybrid of (a) exact/normalised merchant-rule lookup from the user's correction history, (b) keyword/regex knowledge base for Indian merchants (Swiggy→Dining, IRCTC→Travel, BESCOM→Utilities...), (c) on-device TF-IDF + logistic model over payee/note/amount/time features trained from the user's corrections. Precedence: user rule > learned model (if confidence ≥ 0.85) > knowledge base > "Uncategorised" prompt.
- **Stage 2 — Stream classification** (per recurring stream / category): statistical, runs weekly.

8.2 Stream classification algorithm

```
for each expense stream s (merchant-series or category) over last 6 months:
  n      = count(months with activity)
  cv     = stdev(monthly amount) / mean(monthly amount) // coefficient of variation
  cadence = regularity of interval between occurrences (MAD of gaps / median gap)
  dayLock = share of occurrences within  $\pm 3$  days of modal day-of-month

  score = 0.45*(1 - min(cv,1)) + 0.35*(1 - min(cadence,1)) + 0.20*dayLock

  classify:
    score  $\geq 0.75$  and  $n \geq 3$    → FIXED      (rent, EMI, subscriptions, fees)
     $0.45 \leq \text{score} < 0.75$  → SEMI_FIXED (electricity, fuel, groceries)
    otherwise                  → VARIABLE   (dining, shopping, travel)
```

- Known-obligation overrides: transactions linked to loan EMIs, recurring rules, or insurance premia are FIXED regardless of score.
- User can pin any stream's class; pins beat the model and are training signals.
- Output per month: FixedLoad (sum of fixed), SemiFixedExpected (median), VariableBudgetable — stored in monthly_profile (DB) and exposed as evidence.
- Cold start: with < 2 months of data, category-level Indian priors are used and clearly labelled "estimate — improves with data".

8.3 Nature Classification — Need / Want / Investment / Asset / Liability (AI-CLS-N)

Every transaction also gets a **nature** label answering "what did this money become?" — the axis that powers the 50/30/20 view, the true-spend metric, and honest net-worth thinking. Auto-assigned, user-correctable, learned.

Nature	Definition & examples	Effect on metrics
NEED	Essential living costs: rent, groceries, utilities, commute, medicines, insurance premiums, school fees	Counts in true spend; feeds emergency-fund essentials (\$10) and the 50% band
WANT	Discretionary consumption: dining, entertainment, shopping, travel, upgrades	Counts in true spend; the 30% band; highest Purchase-Advisor scrutiny
INVESTMENT	Money that stays yours and grows: SIPs, FD/RD/PPF contributions, goal transfers, stock buys	Excluded from spend; counts as saving rate; the 20% band

Nature	Definition & examples	Effect on metrics
ASSET	Consumption converted to a lasting asset: gold purchase, property payment, vehicle purchase	Excluded from true spend (conversion, not consumption); MUST create/update a linked asset record so net worth stays truthful
LIABILITY	Debt service: EMI, credit-card bill payments, prepayments	Auto-split: principal = liability reduction (conversion, net-worth positive); interest & fees = true cost, counted like NEED

8.3.1 Auto-classification decision order

```

nature(txn):
  1. account-type override: txn on loan account          → LIABILITY
     principal/interest split from the amortisation row
     matching the payment date (loan_amortization_rows)
  2. transfer to investment/goal-linked account          → INVESTMENT
  3. creates/updates an asset record (gold, vehicle,
     property, holdings)                                → ASSET
  4. user rule from past corrections (merchant/category) → learned nature
  5. category default_nature from knowledge base        → NEED or WANT
     (groceries→NEED, dining→WANT, utilities→NEED ...)
  6. context modifier: amount > 3× category median in a
     NEED category → prompt once ("₹9,400 at grocery –
     festival stock-up (NEED) or party (WANT)?")        → remembered
  confidence stored; low-confidence natures flagged for
  one-tap review in the weekly digest, never blocking save

```

8.3.2 Rules & integration

ID	Rule
AI-CLSN-001	transactions gains nature + nature_confidence columns; categories gains default_nature; this subsumes and replaces the category_essentiality table (essential = NEED share).
AI-CLSN-002	The Money and Home screens MUST offer a Need/Want/Invest monthly ring vs the RULE-50-30-20 bands (auto-flexed per §29.2), with each band tappable to its transactions.
AI-CLSN-003	True Spend metric = NEED + WANT + interest/fees. ASSET and INVESTMENT conversions are shown separately ("₹18,000 spent, ₹25,000 converted to assets") so a gold purchase never reads as a bad month.
AI-CLSN-004	EMI principal/interest auto-split posts as two linked lines on the same transaction (split mechanism, FR-TXN-004); the interest line is what budgets and true spend see.
AI-CLSN-005	Nature corrections are training signals with the same learning loop and "apply to similar" flow as FR-TXN-005; user pins beat the model.
AI-CLSN-006	Consumers: 50/30/20 dashboard, FHS spending-discipline pillar (WANT share and trend), emergency-fund essentials, Purchase Advisor (WANT purchases face the full interview ladder §13.3; ASSET/INVESTMENT purchases get the affordability gates but lighter want-scrutiny), monthly ritual summary.

9. Cash-Flow Forecasting Engine (AI-FCT)

9.1 Requirements

ID	Requirement
AI-FCT-001	Produce a daily balance forecast for 90 days (Pro: 12 months) per account and consolidated, with P10/P50/P90 bands.

ID	Requirement
AI-FCT-002	Detect and flag any forecast day where consolidated liquid balance < user buffer (default ₹5,000) — a "crunch day" — and alert per §18.
AI-FCT-003	Forecast MUST decompose into: scheduled items (known EMIs, recurring rules, future-dated txns) + predicted income + predicted variable spend; each component inspectable in UI.
AI-FCT-004	Accuracy tracking: store forecast snapshots, compare with actuals, and surface MAPE to the engine (for confidence) and to the user (honesty).

9.2 Method (deliberately simple, explainable, upgradeable)

```

forecast(day d) = openingBalance
+ Σ scheduledIncome(d)           // salary rule, confirmed recurring credits
+ Σ scheduledOutflow(d)          // EMIs, rent, subscriptions, planned txns
+ predictedVariableSpend(d)      // see below
+ seasonalAdjustment(d)         // §9.3

predictedVariableSpend(d):
  base   = trimmed mean of daily variable spend, last 90 days (trim 10%)
  dowAdj = day-of-week multiplier (weekend vs weekday medians)
  domAdj = day-of-month curve (salary-cycle shape: spike days 1-5, trough 25-31)
  value  = base * dowAdj * domAdj

bands: Monte Carlo (n=500, seeded) resampling daily residuals → P10/P50/P90

```

- Salary detection: largest regular credit with cadence ≈ monthly and dayLock high becomes the income rule; user confirms once.
- Irregular income (Persona 3): income modelled as distribution of monthly totals; P10 income used for conservative planning and the emergency-fund multiplier is raised (§11).
- Upgrade path (Phase 4): replace the heuristic variable-spend model with an on-device gradient-boosted or temporal model behind the same interface — the ForecastEngine contract must not change.

9.3 Seasonality Prediction

- Monthly seasonal index per category = median(month m spend across years) / median(all months), shrunk toward 1.0 when < 2 years of data (index = 1 + k*(raw-1), k = months_observed/24).
- Indian calendar events knowledge base (Diwali, school-admission season, monsoon, wedding season, tax-saving January–March rush) contributes labelled expectations even in year 1.
- Outputs: "October festival spending typically +38% for you — plan ₹6,500 extra" style insights and inputs to forecast/budget suggestions.

10. Emergency Fund Engine (AI-EMF)

10.1 Target calculation

```
monthlyEssentials = FixedLoad + median(SemiFixed) + essentialShare(Variable)
// essential variable = groceries, transport-to-work, medicines...
// category essentiality map is user-editable

multiplier M:
  base = 6
  income volatility (cv of monthly income):
    cv < 0.10 → +0 | 0.10-0.30 → +1 | > 0.30 → +3
  single-earner household with dependents → +1
  no health insurance flag → +1
  job-stability self-assessment (optional) → -1 / 0 / +1
  M clamped to [3, 12]

target = monthlyEssentials * M
runwayMonths = liquidFunds / monthlyEssentials // headline metric
```

- "Liquid funds" = savings + cash + FDs breakable without major penalty + liquid MF; the mapping of account→liquidity tier is stored per account and user-editable.
- Coach behaviour: if runway < 1 month → urgent framing, suggests pausing non-essential goals; 1–3 → build plan from forecast surplus; ≥ target → celebrate, suggest redirecting surplus to goals; > target+2 → suggest excess deployment (with P-07 framing: suggestion only).
- Every number in the explanation links to its evidence (which categories counted as essential, which accounts counted as liquid).

11. Investment Intelligence Module (AI-INV)

11.1 Scope & guardrails

RULE / NOTE: The module analyses, educates, and flags — it does not recommend specific securities or funds. Outputs are framed as observations with options, consistent with P-07 and Indian regulatory caution (not SEBI-registered advice). This wording **MUST** appear in the module footer.

11.2 Capabilities

Capability	Definition
Allocation analysis	Actual allocation across equity / debt / gold / real-estate / cash / crypto vs a target band derived from user's selected risk profile (Conservative/Balanced/Growth presets, editable)
Diversification score	0–100 from: asset-class spread (Herfindahl index), single-holding concentration (> 15% flag), single-asset-class concentration (> 70% flag), overlap of goals on same asset
XIRR & benchmarking	Money-weighted return per holding and portfolio; optional comparison to cached index series when online
Rebalancing signal	Drift alert when any class deviates > 5 pp from target band; shows the delta amounts, never auto-executes
SIP consistency	Detects missed SIP months from transaction stream; quantifies long-term cost of the gap
Idle-cash detector	Savings balance persistently > (STS needs + emergency buffer) for 60 days → education card on deployment options
Crypto & gold sanity	Volatile-asset share > user risk cap → flag; gold valued daily from cached price feed (§17)

12. Vehicle & Appliance Maintenance Prediction (AI-VEH)

12.1 Inputs

- Odometer log (manual entries + fuel-log odometer readings).
- Service history (date, odo, type, cost) and a bundled maintenance knowledge base: per vehicle class (2W/hatch/sedan/SUV/EV), service intervals in km and months, typical Indian cost ranges, consumables (tyres ~40k km, battery ~4 yr, brake pads, insurance/PUC renewals).
- Fuel log for mileage trend; usage rate (km/month) estimated by robust regression over odometer readings.

12.2 Prediction logic

```

kmPerMonth      = robustSlope(odometer readings, last 12 months)
nextService.dueAt = min( lastService.odo + interval_km → date via kmPerMonth,
                        lastService.date + interval_months )
predictedCost    = knowledgeBase.costRange(vehicleClass, serviceType)
                  adjusted by user's own service-cost history (personal index)

alerts: 30 days / 500 km before due → "Service due ~Aug 20, expect ₹3,000-4,500"
        mileage drop > 15% vs 3-month baseline → "Possible maintenance issue"
        insurance/PUC expiry -30/-7 days → renewal reminders with amount

Forecast integration: predicted service costs are injected into the cash-flow
forecast as scheduled outflows with confidence bands (AI-FCT-003 components).

```

- Appliances (Phase 4): same engine, different knowledge base (AC service pre-summer, water-purifier filters, extended-warranty expiry).

13. Purchase Advisor Decision Engine (AI-PA)

The flagship feature. Input: item, price, payment method, urgency, optional category. Output: a verdict with a fully transparent reasoning trace.

13.1 Decision pipeline

```

1. AFFORDABILITY – can cash/limit absorb it without breaching buffers?
   liquidAfter = liquidFunds – price (or EMI impact on monthly surplus)
   fail if liquidAfter < emergencyFloor → verdict ≤ STRETCH

2. CASH-FLOW IMPACT – re-run 90-day forecast including the purchase;
   count new crunch days; any crunch day → downgrade verdict

3. OBLIGATION RATIO – (EMIs + rent + new EMI) / income after purchase;
   > 50% hard flag, > 40% warning (Indian lender heuristics)

4. GOAL IMPACT – recompute goal ETAs; report per-goal delay in days

5. BUDGET FIT – category budget remaining this month & 3-month trend

6. OPPORTUNITY COST – value if invested instead:
   FV = price × (1+r)^t for t ∈ {5y, 10y}, r = user assumption (default 11% eq.)

7. TIMING – seasonality knowledge base (sale seasons), predicted crunch/
   surplus months → "better in March" style suggestion

VERDICT = worst severity across gates, softened by urgency flag:
COMFORTABLE | STRETCH (with conditions) | NOT NOW (with revisit date)

```

13.2 Output contract (rendered as the reasoning card)

- Verdict + one-line summary.
- Gate-by-gate table: each check, its numbers, pass/warn/fail.
- Impact strip: liquid funds before/after, runway before/after, goal delays.
- Alternatives: wait N months (computed date when verdict flips to Comfortable), cheaper threshold price, EMI vs cash comparison with total interest.
- One-tap actions: "Create saving goal for this", "Remind me on <date>", "Ignore".
- The full trace object is persisted so the user can revisit why a past decision was made (AI-ARC-006).

13.3 Buy List & Adaptive Purchase Interview (AI-PA-INT)

Users add wishes ("headphones ₹8k", "mobile ₹45k") to a **Buy List** instead of buying impulsively. On add — and again before purchase — the AI runs a short interview whose **depth scales with the purchase's weight in the user's finances**, then either strengthens the wish, suggests an alternative, or recommends removing it (user always confirms; Control pillar).

13.3.1 Question escalation ladder (price relative to earning & spending)

Purchase weight (price ÷ monthly income, or EMI ÷ monthly surplus)	Questions	Interview content
< 0.5% (casual)	0–1	No friction; at most one tap: "Need or want?"
0.5–2% (small)	2–3	+ "How often will you use it?" + duplicate check against owned items

Purchase weight (price ÷ monthly income, or EMI ÷ monthly surplus)	Questions	Interview content
2–10% (significant)	4–5	+ "What breaks if you wait 30 days?" + cheaper-alternative prompt + which goal it delays (computed, shown)
10–25% (major)	6–7	+ Worth-It history for the category (§32) + total cost of ownership (accessories, subscriptions, maintenance) + timing signal (sale season / crunch forecast)
> 25% or any EMI (heavy)	Full interview 8+	All of the above + full Purchase Advisor gates (§13.1) + 24h mandatory cooling-off before "Buy now" unlocks (RULE-COOL-OFF escalated)

13.3.2 Interview logic & outcomes

question engine: bank of typed questions, each mapping answers → score deltas
 WantScore(item) starts at 50, adjusted by:
 need vs want, expected uses/month, duplicate-ownership hit,
 goal-delay tolerance answer, Worth-It category score (§32),
 urgency claim vs 30-day cooling behaviour (still wanted later = +)

outcomes:

WantScore ≥ 70 → KEEP, watch for buy timing (PA verdict + sale season)
 40–69 → PARK: stays on list, auto re-interview in 30 days;
 "still want it" twice = organic promotion
 < 40 → SUGGEST REMOVE, with the user's own answers as evidence
 ("you said 2 uses/month and you own X that does this");
 one tap removes, one tap overrides – never auto-deleted

- **Alternative finder:** before any buy verdict, the engine searches what the user already owns (assets, past purchases in the category via merchants/attachments) — "You bought Sony WH-CH520 14 months ago, rated Worth It; what's missing from it?" — and offers a lower-tier price point ("a ₹4k option covers your stated 3 uses/month; the ₹8k tier pays off above 10 uses").
- **Answers are stored, not wasted:** interview responses feed the personal value model (§32) and the Purchase Advisor trace, so the app's questions get sharper and fewer over time — a well-understood user gets a shorter interview (EASY pillar).
- **Buy-timing watch:** kept items are monitored: PA verdict recomputed on payday/crunch changes, sale-season alerts from the calendar KB, and optional target-price alert; the notification says "your ₹45k mobile: Comfortable this month, and Diwali sales start in 12 days."
- **Honesty rule:** the interview may challenge, never nag — max one re-interview per item per 30 days, tone per WRT-002, and an explicit "stop asking about this item" option.
- New tables: wishlist_items(id, name, est_price_minor, category_id, urgency, want_score, status[watching/parked/removed/bought], target_price_minor NULL), interview_sessions(item_id, asked_json, answers_json, score_delta, created_at). Chat tools added: add_to_buylist(item, price), review_buylist().

14. Financial Health Score (AI-FHS)

14.1 Methodology — 0 to 1000, five weighted pillars

Pillar	Weight	Scoring basis (each pillar scored 0–100 then weighted)
Liquidity & Emergency	25%	Runway months vs personal target M (\$10): 100 at $\geq M$, linear below; floor boost for any runway ≥ 1 month
Debt Health	20%	Obligation ratio (EMI+rent)/income ($\leq 30\% \rightarrow 100$, $\geq 55\% \rightarrow 0$), credit-card utilisation ($< 30\%$ good), revolving-interest flag, no-debt bonus
Spending Discipline	20%	Savings rate (saved/income, 100 at $\geq 30\%$), budget adherence %, overspend-alert frequency, spend volatility (cv)
Goals & Future	20%	On-track share of goal portfolio, retirement contribution presence, goal funding consistency streak
Protection & Growth	15%	Insurance flags (health/term self-declared), diversification score (\$11), idle-cash penalty, single-income-source penalty

- Score bands: 800+ Excellent, 650–799 Good, 500–649 Fair, 350–499 Needs Attention, < 350 At Risk.
- **Anti-anxiety rules:** score updates at most weekly; movement shown with cause list ("–12: credit-card utilisation rose to 46%"); every drop pairs with the single highest-leverage action; no red flashing, no gamified shame.
- Each pillar drill-down shows the exact inputs, formula, and what-if slider ("if utilisation drops to 30%, +9 points").
- Insufficient-data handling: pillars with < 1 month of signal render as "—" and are re-weighted, never guessed.

15. Goal Planning Engine (AI-GOAL)

15.1 Core math

```

requiredMonthly (with expected annual return r, n months to target):
  i = (1+r)^(1/12) - 1
  PMT = (target - current*(1+i)^n) * i / ((1+i)^n - 1)    // future-value annuity
  r defaults by funding bucket: savings 3%, FD/debt 6.5%, hybrid 9%, equity 11%
  inflation-adjusted targets: target * (1+inf)^(years) for education/retirement

feasibility:  $\sum \text{requiredMonthly}(\text{all active goals}) \leq \text{P50 forecast surplus}$ 
  if infeasible  $\rightarrow$  gap analysis + 3 levers per goal (date / amount / contribution)

retirement corpus (simplified, editable assumptions):
  corpus = annualExpensesAtRetirement / SWR    // SWR default 4%, editable

```

- Priority waterfall: Emergency Fund $\rightarrow$ high-priority goals $\rightarrow$ others; monthly surplus allocation suggestion follows the waterfall and is shown as a draggable plan.
- Progress is transaction-evidenced (FR-GOAL-004); "ghost progress" (manual claims without linked funds) is visually distinct.
- Monthly goal digest: on-track/off-track per goal with the one action that fixes the worst one.

16. External Data Integration (AI-EXT)

External data enriches valuation and advice. All of it is **optional, cached, and clearly time-stamped**; the app is fully functional without it (P-04).

16.1 Data feeds

Feed	Used by	Refresh	Source strategy
CPI inflation (India)	Goal inflation adjustment, real-return displays	Monthly	Own backend proxy over MoSPI/public series
RBI repo & FD rate ranges	Loan-vs-invest simulator, idle-cash education, EMI context	On change / weekly	Backend proxy over RBI publications
Fuel prices (city-wise petrol/diesel)	Vehicle cost tracking, commute insights	Daily	Backend proxy over public price feeds
Gold & silver prices (INR)	Gold asset revaluation	Daily	Backend proxy
Index & MF NAV series	Portfolio valuation, benchmark context	Daily EOD	AMFI NAV feed via backend proxy
Crypto spot prices	Crypto asset revaluation	15 min when app open	Backend proxy
Index valuation: P/E, P/B, dividend yield (NSE indices)	AI-MKT SIG-VALUATION, yield-gap context	Daily EOD	Backend proxy over NSE published data
India VIX	AI-MKT SIG-VIX fear signal	Daily EOD	Backend proxy over NSE
10-yr G-Sec yield	Yield-gap context, prepay-vs-invest simulator	Daily	Backend proxy
Index/instrument OHLC history	AI-MKT signal computation & backtests	Daily EOD + 5y on first watch	Backend proxy
Calendar/festival knowledge base	Seasonality engine	Bundled + yearly update	Ships in app

16.2 Rules

ID	Rule
EXT-001	The client never calls third-party sources directly; all feeds go through the app's own read-only Market Data API (§22) so sources can change without app releases and no third party sees user IPs per-feature.
EXT-002	Every cached datum stores fetchedAt + TTL; UI shows "as of <date>" when stale, and engines widen confidence bands when inputs are stale.
EXT-003	Requests contain zero personal financial data — only instrument identifiers and city (for fuel, coarse, user-chosen).
EXT-004	Feed failure degrades to last cache; engines must never crash or block on network.

17. Notification Engine (AI-NTF)

17.1 Notification taxonomy

Type	Examples	Default
Critical money events	Crunch-day warning, bill due tomorrow, card due with amount	ON, can bypass quiet hours (user choice)
Budget & discipline	80%/100% budget alerts, pace warnings	ON
AI insights	Weekly digest, seasonal warnings, idle-cash, health-score change	Weekly digest ON, per-insight OFF
Maintenance & renewals	Vehicle service, insurance, PUC, subscription renewal price hikes	ON
Habit & capture	Daily capture reminder (if enabled), uncategorised backlog	OFF

Type	Examples	Default
Goal events	Milestones, off-track, contribution reminder on salary day	ON

17.2 Policy rules

ID	Rule
NTF-001	Global budget: max 2 non-critical notifications/day and 8/week; excess folds into the weekly digest. Critical events are exempt.
NTF-002	Quiet hours (default 22:00–08:00) delay non-critical delivery; batching prefers the user's observed app-open hour.
NTF-003	Every notification deep-links to its evidence screen and carries inline actions (Mark paid / Snooze / View).
NTF-004	Notification content on lock screen MUST respect a "hide amounts" setting (show "A bill is due tomorrow" instead of amount).
NTF-005	All notifications are generated locally (no push infrastructure needed for v1); WorkManager schedules evaluation jobs.
NTF-006	Each Android channel maps 1:1 to a taxonomy row so OS-level control matches in-app settings.

18. OCR & Receipt Scanning Workflow (detail)

18.1 Pipeline

```
capture → preprocess (crop, skew, contrast) → ML Kit text recognition (on-device)
  → layout heuristics: candidate totals = currency-formatted numbers;
    total = max plausible candidate near keywords {total, grand, amount, payable}
  → date parser (dd/mm/yy variants, Indian formats) → merchant = top-region text
    matched against merchant KB + user history
  → GST heuristic: lines with {GST, CGST, SGST, %} → tax field
  → line items (best effort): qty-name-price table detection
  → confidence per field → review screen → save transaction + encrypted image
```

- Targets: total-amount accuracy ≥ 95% on printed receipts, ≥ 80% field-complete extraction; review screen makes 100% correctness a two-tap job.
- All failures fall back to manual entry pre-filled with whatever was extracted.
- Images stored in app-private encrypted storage; EXIF stripped; auto-compress > 1 MB.

19. AI Chat Assistant Design (AI-CHAT)

19.1 Principles

- The assistant is a **narrow financial copilot over the user's own data**, not a general chatbot.
- Architecture: intent parsing → **tool calls into the engines** → structured results → LLM verbalisation → guardrail check (AI-ARC-004) → render with evidence chips.
- The LLM never sees raw full transaction history; tools return aggregates and top-k relevant records only.

19.2 Tool registry (the only way chat touches data)

Tool	Contract
query_spend(period, category?, merchant?)	Aggregated totals, counts, top merchants
get_balance(accountId? liquid)	Current balances and runway
get_forecast(days)	Forecast summary + crunch days
get_budget_status(month)	Per-budget spent/remaining/projection
get_goals()	Goal list with ETA and on-track state
purchase_check(item, price, method)	Full Purchase Advisor verdict (§13)
get_health_score()	FHS with pillar breakdown
explain_insight(insightId)	Evidence pack for a shown insight
simulate(scenario)	What-if: income change, new EMI, rate change, expense cut
create_txn_draft(fields)	Prefilled add-transaction screen — user must confirm; chat never writes directly
market_status(instrument) / opportunity_check(instrument)	AI-MKT verdict with signal trace and backtest line (§30)
add_to_buylist(item, price) / review_buylist()	Buy List with adaptive interview entry (§13.3)
convert_currency(amount, from, to)	FX conversion using cached daily rates (§34.2)

19.3 Behaviour rules

ID	Rule
CHT-001	Every numeric statement in a reply MUST originate from a tool result; the guardrail cross-checks and blocks fabricated figures.
CHT-002	Out-of-scope requests (stock tips, tax filing, legal) get a friendly refusal + what the assistant CAN do.
CHT-003	Suggested chips ("Where did money go this month?", "Can I afford ₹X?", "When can I buy the bike?") drive discovery; free text also supported.
CHT-004	Conversations are stored locally, deletable, and excluded from backup by default.
CHT-005	On-device model is the default engine; if the user enables cloud assist, the consent screen lists exactly the structured fields that will be sent, and every cloud call is logged in the consents dashboard.

20. Database Schema (Room / SQLCipher)

Single source of truth. Conventions: snake_case table names; TEXT UUID primary keys (id); all money as INTEGER minor units (paise) + currency code; all timestamps as epoch millis UTC (INTEGER) with named exceptions for date-only fields (ISO yyyy-mm-dd TEXT); soft delete via deleted_at; created_at/updated_at on every table; foreign keys ON with indices on every FK; schema versioned with tested Room migrations from v1 forever.

20.1 Table inventory (by domain)

Domain	Tables
Identity & profile	profiles, household_members, user_settings, consents, feature_flags
Accounts & assets	accounts, account_balances_history, credit_card_details, loan_details, loan_amortization_rows, holdings, holdingLots, asset_valuations, gold_items, properties, vehicles
Transactions	transactions, transaction_splits, transaction_tags, tags, attachments, categories, category_essentiality, merchants, merchant_aliases, transfers
Recurring & scheduled	recurring_rules, recurring_occurrences, scheduled_items, subscriptions
Budgets & goals	budgets, budget_periods, goals, goal_contributions, goal_priority_plan
Vehicle ops	odometer_logs, fuel_logs, service_records, service_knowledge_base, renewal_items
AI & analytics	monthly_profiles, category_stats, seasonal_indices, forecasts, forecast_points, forecast_accuracy, insights, insight_feedback, decisions (purchase-advisor traces), health_scores, health_score_components, classification_rules, model_metadata, chat_sessions, chat_messages, chat_tool_calls
External data	market_instruments, price_points, macro_indicators (CPI, repo), fuel_prices, calendar_events
Rulebook, signals & advisor	rules_knowledge_base, watchlist, price_series, signal_configs, signal_verdicts, backtest_results, wishlist_items, interview_sessions, usage_logs, purchase_reviews, value_model_scores, fx_rates
System	notifications_log, jobs_log, audit_log, backups_log, schema_migrations_log, import_batches, export_batches, sms_parse_rules (opt-in), kv_cache

Total: ~75 tables at v1.4 of this SRS; the schema is deliberately normalised so the "100+ tables" ceiling is available for Phase 5 additions (appliances, insurance, tax detail, household ledgers) without redesign.

20.2 Core table definitions (abridged DDL)

```
accounts(id PK, profile_id FK, type TEXT CHECK(type IN ('bank','cash','credit_card',
'loan','investment','gold','crypto','property','vehicle','receivable','payable')),
name, institution, currency, opening_balance_minor INT, current_balance_minor INT,
liquidity_tier INT, -- 0 instant / 1 breakable / 2 locked / 3 illiquid
include_in_networth INT, archived_at, created_at, updated_at, deleted_at)

transactions(id PK, profile_id FK, account_id FK, type TEXT CHECK(type IN
('expense','income','transfer_out','transfer_in','adjustment')),
amount_minor INT NOT NULL, currency TEXT NOT NULL, occurred_at INT NOT NULL,
category_id FK, merchant_id FK, payee TEXT, note TEXT,
transfer_id FK NULL, recurring_rule_id FK NULL,
source TEXT CHECK(source IN ('manual','ocr','sms','import','recurring')),
classification TEXT NULL, -- fixed / semi_fixed / variable (stream result)
cls_confidence REAL,
nature TEXT NULL, -- need/want/investment/asset/liability (§8.3)
nature_confidence REAL, is_future INT DEFAULT 0,
created_at, updated_at, deleted_at)
INDEX(account_id, occurred_at), INDEX(category_id, occurred_at), INDEX(merchant_id)
```

```

recurring_rules(id PK, profile_id, name, direction, amount_minor, tolerance_pct,
  cadence TEXT,          -- rrule-like: MONTHLY;BYMONTHDAY=5 ...
  account_id, category_id, merchant_id, next_due_date TEXT, autopost INT,
  end_date TEXT NULL, source TEXT, created_at, updated_at)

goals(id PK, profile_id, type, name, target_amount_minor, current_amount_minor,
  target_date TEXT NULL, priority INT, expected_return_bps INT,
  inflation_adjust INT, status TEXT, created_at, updated_at)

insights(id PK, profile_id, engine_id, engine_version, type, fingerprint UNIQUE,
  severity, title, body, evidence_json, action_json NULL, confidence REAL,
  valid_from, valid_until, status TEXT,  -- active/dismissed/snoozed/acted
  created_at, updated_at)

forecasts(id PK, profile_id, horizon_days, generated_at, engine_version,
  assumptions_json) → forecast_points(forecast_id FK, day TEXT,
  p10_minor, p50_minor, p90_minor, scheduled_minor, predicted_minor)

health_scores(id PK, profile_id, computed_at, total INT,
  band TEXT) → health_score_components(score_id FK, pillar, raw REAL,
  weighted REAL, inputs_json, delta_reason TEXT)

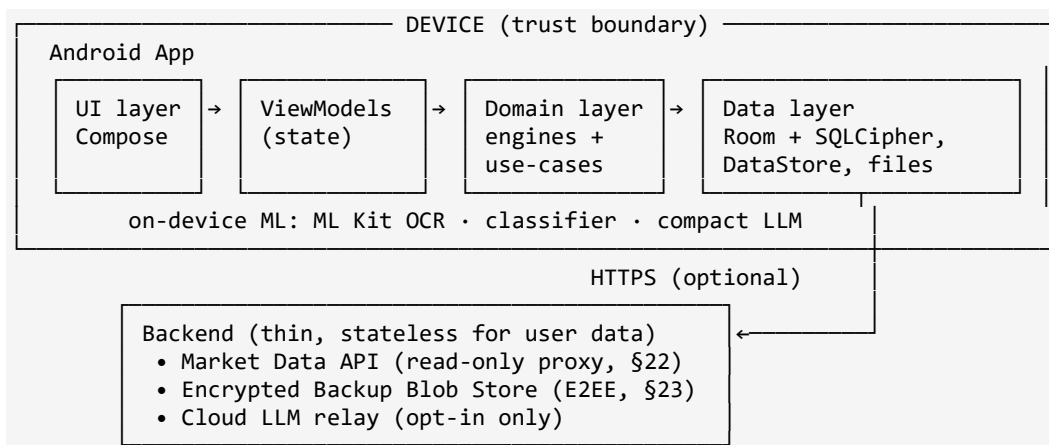
```

20.3 Data-layer rules

ID	Rule
DB-001	Balances are never mutated ad hoc: current_balance is derivable from opening balance + transactions and is verified by a nightly integrity job; discrepancies raise an internal alert and an adjustment prompt (FR-ACC-006).
DB-002	Deletes are soft (deleted_at) everywhere user data is involved; hard purge only via the erase-all flow and the 30-day trash compactor.
DB-003	Every schema change ships with a Room Migration + migration test using a real pre-change database fixture. Destructive migrations are forbidden.
DB-004	All multi-row operations (transfer, split, import batch) run in a single transaction.
DB-005	evidence_json / inputs_json blobs are versioned ({"v":1,...}) so old insights render after upgrades.
DB-006	Attachments live on the encrypted filesystem, referenced by content hash; the DB stores metadata only.

21. System & Android App Architecture

21.1 System context



- The backend never stores plaintext user financial data. Backup blobs are client-side encrypted; the server cannot read them (§23.3).
- The app must build and pass all tests with the backend absent (P-04). Backend outage = market data goes stale, backup queues.

21.2 Module structure (Gradle multi-module)

```

:app                - DI graph assembly, navigation host, MainActivity
:core:model         - pure Kotlin data classes, Money, Ids (no deps)
:core:common        - Result/AppError, time (Clock), dispatchers, logging
:core:database      - Room entities, DAOs, migrations, SQLCipher setup
:core:datastore     - settings & consents (Proto DataStore)
:core:network       - Retrofit/OkHttp, Market Data API client, DTOs
:core:crypto        - key management, Keystore, encryption helpers
:core:designsystem   - theme, tokens, shared composables, charts
:domain:engines:*   - one module per engine: classification, forecast,
                    emergencyfund, investment, vehicle, purchase,
                    healthscore, goals, seasonality, rules, orchestrator,
                    marketsignal, worthit, growth, anomaly
:domain:usecase     - thin use-cases composing repositories + engines
:data:repository    - repository implementations (offline-first)
:ml:ocr             - receipt pipeline over ML Kit
:ml:llm             - LLM abstraction (on-device impl + cloud impl)
:feature:home       :feature:transactions :feature:budgets :feature:goals
:feature:accounts   :feature:investments :feature:vehicle :feature:advisor
:feature:chat       :feature:onboarding  :feature:settings :feature:scan
:sync:backup        - E2EE backup/restore workers
:widget            - Glance home-screen widget

```

ID	Architecture rule
ARC-001	Dependency direction is one-way: feature → domain → data/core. Feature modules never depend on each other; cross-feature navigation goes through the nav graph with typed routes.
ARC-002	:domain:* and :core:model are pure Kotlin (JVM) modules — no Android imports. This is enforced by Gradle (no android plugin) and keeps engines portable for future KMP/iOS.
ARC-003	Every engine exposes exactly one public interface + result types; implementations are internal and injected via Hilt.
ARC-004	UI state is a single immutable data class per screen (UIState) exposed as StateFlow; events flow up via a sealed interface. No LiveData, no mutable state leaking from ViewModels.
ARC-005	Repositories are the only classes that touch DAOs or network. ViewModels never see Room/Retrofit types — only domain models.
ARC-006	All async work uses structured concurrency (viewModelScope / injected scopes). GlobalScope is banned by lint.

21.3 Technology stack (pinned decisions)

Concern	Choice	Rule / rationale
Language	Kotlin (100%), JDK 17 toolchain	No Java sources
UI	Jetpack Compose + Material 3	No XML layouts except widget/notification remoteviews
Navigation	Navigation-Compose, type-safe routes	One nav graph per feature, assembled in :app
DI	Hilt	Constructor injection only; no service locators
Persistence	Room + SQLCipher (AES-256)	WAL mode; key in Android Keystore (§23)

Concern	Choice	Rule / rationale
Preferences	Proto DataStore	SharedPreferences banned
Background	WorkManager	All periodic jobs; expedited only for backup-on-demand
Network	Retrofit + OkHttp + kotlinx.serialization	Certificate pinning to own backend
Images/OCR	CameraX + ML Kit Text Recognition v2	On-device only
On-device LLM	AICore / MediaPipe LLM (Gemma-class), behind LlmEngine interface	Swappable; cloud impl opt-in
Charts	Custom Compose canvas charts in :core:designsystem	No heavyweight chart lib
Testing	JUnit5, Turbine, MockK, Robolectric, Compose UI tests, Paparazzi screenshots	See §21.5
Static analysis	ktlint + detekt + Android lint + custom lint (Money, GlobalScope, string literals)	CI-blocking
CI/CD	GitHub Actions: build, test, lint, screenshot diff, assemble	main is always releasable

21.4 Money & time rules (the two classic bug factories)

```

@JvmInline value class Money(val minor: Long) {           // paise
    operator fun plus(o: Money) = Money(minor + o.minor)
    operator fun minus(o: Money) = Money(minor - o.minor)
    fun percentOf(bps: Int) = Money(Math.multiplyHigh-safe rounding ...)
}
// Rendering only at the UI edge: MoneyFormatter(profile.localeStyle)

```

ID	Rule
MNY-001	Money is Long minor units end-to-end (DB → engine → UI). Double/Float on money fails CI via custom lint. Division uses explicit rounding mode (HALF_EVEN) and remainder distribution for splits.
MNY-002	Percentages and rates are integer basis points (bps) in engines; conversion helpers live in :core:model only.
TIM-001	All timestamps stored UTC epoch millis; all calendar logic (day rollover, month boundaries, due dates) computed in the profile time zone via injected Clock — never System.currentTimeMillis() in domain code (lint-banned).
TIM-002	Date-only fields (due dates, goal dates) are ISO LocalDate strings; never midnight timestamps.

21.5 Testing strategy

Level	Policy
Engine unit tests	Every engine: golden-file tests (fixed input snapshot → expected result JSON), property tests for math (splits always sum, forecast monotonic identities), and seeded-random determinism tests. Coverage ≥ 85%, money math 100%.
Repository tests	Room in-memory + migration tests against real schema fixtures for every version bump.
ViewModel tests	Turbine on StateFlow: given repository fakes, assert full UiState sequences including loading/error.
UI tests	Compose tests for critical flows (add txn ≤ 3 taps, onboarding, purchase advisor); Paparazzi screenshot tests for design-system components in light/dark/200% font.
E2E smoke	One instrumented run per release: onboard → add data → verify dashboard, forecast, export/import round-trip, airplane-mode pass.

Level	Policy
AI evaluation	Frozen labelled datasets: 1k categorisation samples (target $\geq 92\%$), 200 receipt images (total $\geq 95\%$), forecast backtests on synthetic personas; regression thresholds block merges.

21.6 Coding standards (binding checklist)

- **Style:** ktlint official style; max function ~ 40 lines guided by detekt; no wildcard imports; files organised by feature not by type.
- **Nullability:** platform types isolated at the network/DB edge; domain models are null-hostile — model absence with sealed types, not null.
- **Errors:** no exceptions across layer boundaries; repositories/engines return `Result<T, AppError>` (sealed). Crashes only for programmer errors.
- **Immutability:** data classes with `val` everywhere; collections exposed as read-only.
- **Concurrency:** dispatchers injected (`DispatcherProvider`); DAOs are `suspend/Flow`; no `runBlocking` outside tests.
- **Logging:** structured logger with levels; PII/amount logging banned in release builds (lint rule strips it); `audit_log` table for security events.
- **Strings & resources:** every user-visible string in `strings.xml` with ICU plurals; every colour/dimension from theme tokens.
- **Commits & branches:** trunk-based; conventional commits (`feat/fix/refactor...`); each commit references requirement IDs; feature flags for incomplete work — no long-lived branches.
- **Reviews:** PR template includes DoD checklist (§4.2); AI-generated code is reviewed to the same standard as human code.
- **Docs:** every engine module has an `ENGINE.md` (contract, formula, assumptions, version log). Architecture Decision Records (ADRs) for any deviation from this SRS.

RULE / NOTE: Rule of thumb for all decisions not covered here: prefer the boring, testable, explainable option. Cleverness is a cost.

22. API Design (Backend, minimal by design)

22.1 Principles

- REST + JSON, versioned path prefix /v1; additive evolution only within a version.
- No user financial data endpoints exist except opaque encrypted backup blobs.
- Auth: anonymous device key-pair registration (no email required); optional account linkage only for multi-device backup. TLS 1.3 + certificate pinning.
- Idempotency keys on all PUT/POST; standard error envelope {code, message, retryable}.

22.2 Endpoints

Endpoint	Method	Purpose / notes
/v1/market/prices?ids=...	GET	Batch quotes: gold, silver, crypto, indices; ETag caching
/v1/market/nav?schemes=...	GET	MF NAVs (AMFI-sourced), EOD
/v1/market/macro	GET	CPI series, repo rate, small-savings/FD rate ranges
/v1/market/fuel?city=...	GET	Daily petrol/diesel price for a coarse city id
/v1/market/history?ids=...&days=...	GET	Daily OHLC series for AI-MKT signals & backtests (5y on first watch)
/v1/market/fx?pairs=...	GET	Daily FX reference rates for multi-currency consolidation (§34.2)
/v1/knowledge/tax?fy=...	GET	Versioned Indian tax rule tables for Tax Optimizer Lite (§33.1)
/v1/knowledge/calendar?year=...	GET	Festival/season events for seasonality engine
/v1/knowledge/maintenance?class=...	GET	Vehicle maintenance KB updates
/v1/backup/blob	PUT/GET/DELETE	E2EE backup blob (≤ 50 MB), server-blind; version header for conflict detection
/v1/llm/chat	POST	Opt-in cloud LLM relay; body = structured context pack (§19), zero raw txns; no server-side retention
/v1/device/register	POST	Anonymous device registration → signed token

ID	Rule
API-001	Client treats every endpoint as optional: timeouts 5 s, exponential backoff, circuit breaker, silent degradation to cache (EXT-004).
API-002	Rate limits enforced server-side per device token; client batches (one market refresh per app-open, WorkManager daily job otherwise).
API-003	Backup blobs carry a client-side manifest (schema version, created_at, device) inside the ciphertext; the server sees size and version header only.
API-004	The /v1/llm/chat relay logs no bodies, stores no conversations, and is deployed in an Indian region for data-residency expectations.

23. Privacy & Security Model

23.1 Threat model (what we defend against)

Threat	Mitigation
Lost/stolen unlocked phone	App-level lock (biometric/PIN) with auto-lock timer (default 1 min), privacy blur, hide-amounts notifications
Lost/stolen locked phone / extracted storage	SQLCipher AES-256 database + encrypted attachments; key wrapped in hardware-backed Android Keystore (StrongBox where available)
Malicious apps / screen capture	FLAG_SECURE on all screens (settings toggle for user screenshots), no sensitive data in recents thumbnail
Server breach	Nothing to steal: backups are client-side encrypted (server-blind); market API holds no user data
Network attacker	TLS 1.3, certificate pinning, no cleartext traffic (manifest-enforced)
Shoulder surfing	Privacy blur toggle, masked amounts option, discreet app-icon option (MAY, Phase 4)
Data-hungry vendor (us)	Analytics default OFF, aggregate-only, consents dashboard with one-tap revoke, open data-export

23.2 Security requirements

ID	Requirement
SEC-001	Database key: 256-bit random, generated on first run, wrapped by a Keystore key requiring user authentication; never written to disk unwrapped, never leaves the device.
SEC-002	Biometric unlock via BiometricPrompt (class 3) with PIN fallback; 5 failed PIN attempts → 30 s lockout doubling.
SEC-003	All crypto via Google Tink / platform APIs — no hand-rolled crypto (review-blocking rule).
SEC-004	SMS parsing (opt-in) reads only sender-whitelisted bank/UPI templates, processes on-device, stores parsed fields only, and never uploads SMS content.
SEC-005	Backups: AES-256-GCM with a key derived from a user passphrase via Argon2id; the recovery phrase screen makes irrecoverability explicit. Local backup to user-chosen storage (SAF) supported for cloud-averse users.
SEC-006	Erase-all destroys the wrapped DB key first (crypto-shredding), then files, then reports completion.
SEC-007	Dependencies scanned in CI (OSV); release builds are minified, with debuggable/backup flags off and Play integrity checks on.
SEC-008	Compliance posture: DPDP Act 2023 aligned — purpose-limited consent, data-principal rights served locally (export = access right, erase-all = erasure right), no processing of children's data (18+ self-declaration).

24. UI/UX Guidelines

24.1 Design language

- **Material 3, dynamic colour aware**, with a fixed brand palette fallback: deep teal primary, warm neutral surfaces; semantic colours: green = inflow/on-track, amber = warning, red = overspend/risk (never colour alone — icons + labels, §3.5).
- **Typography**: one type scale from Material 3; tabular figures for all amounts; Indian digit grouping (₹1,23,456.78).
- **Numbers-first hierarchy**: the number is the headline, the label is secondary; deltas always show direction + period ("↑ ₹2,340 vs June").

- **Tone of voice:** calm, factual, never shaming ("Dining is 40% over budget" not "You overspent again!"); AI text always evidence-linked; no dark patterns, no fake urgency.
- **Motion:** subtle, functional (state changes, chart draw-in ≤ 300 ms); reduced-motion setting respected.
- **Empty states teach:** every empty screen shows a one-line value promise + primary action + "try with sample data".
- **Dark mode:** first-class, AMOLED-friendly true-black option.

24.2 Interaction rules

ID	Rule
UX-001	Destructive actions get undo (snackbar, 5 s) rather than confirmation dialogs, except irreversible ones (erase-all) which get typed confirmation.
UX-002	Every AI number is tappable → evidence sheet (inputs, formula name, period). "Why?" is a universal affordance on insight cards.
UX-003	Loading: skeletons over spinners; all local screens must render instantly from cache and never show a blocking loader.
UX-004	Forms validate inline on blur; the save button is always visible (sticky) and disabled states explain why.
UX-005	One primary action per screen; FAB is reserved for "add transaction" globally.
UX-006	Haptics: light tick on save/success; none on errors (visual + text instead).

25. Screen-by-Screen Wireframe Notes & User Flows

25.1 Wireframe notes (textual blueprints for hi-fi design)

Screen	Layout blueprint (top → bottom)
Home	Greeting + privacy eye • Net-worth card (amount, 30-day sparkline, Δ) • Safe-to-Spend hero card with "why" link • Month progress ring (spent vs budget) • Upcoming obligations horizontal list (next 14 days, amounts, due chips) • Insight carousel (max 3) • Goal chips row • bottom nav
Transactions	Search bar + filter chips • sticky month header with totals • day-grouped list (icon, payee, category chip, amount, source glyph) • FAB add • multi-select app-bar on long-press
Add transaction	Numeric keypad-first amount field • expense/income/transfer segmented control • smart category suggestion row (top-3 predicted) • account, date (default today), payee autocomplete • expanders: splits, tags, note, attachment • sticky Save
Receipt review	Image top (zoomable) • extracted fields list with confidence dots • per-field edit • duplicate-merge banner when triggered • Save
Budgets	Month switcher • overall bar (spent/remaining/projected) • per-budget rows: category, mini progress bar, pace indicator (ahead/on/behind), projected EOM • "Suggest budgets" action
Goals	Priority-ordered cards: name, progress ring, ETA, on-track badge, required monthly vs actual • add goal • conflict banner when over-committed
Accounts	Net-worth header + chart with range tabs • sections by type with subtotal • account rows: balance, institution, staleness dot for manual assets • reconcile action in overflow
Investments	Portfolio value + XIRR header • allocation donut vs target band overlay • diversification score chip → detail • holdings list with Δ • rebalancing banner when drifted
Vehicle	Vehicle card (photo, odo, monthly cost) • next-service prediction card (date, km, cost range) • renewals timeline (insurance/PUC) • fuel-log quick add + mileage trend chart

Screen	Layout blueprint (top → bottom)
Advisor hub	FHS gauge + band + Δ with reason • pillar chips → drill-down • insight feed (act/snooze/dismiss) • tool grid: Purchase Advisor, Simulators, Chat
Purchase Advisor	Input sheet (item, price, method, urgency) → verdict screen: verdict banner, gate table (pass/warn/fail rows), impact strip (before/after), alternatives cards, action buttons
Chat	Message list with evidence chips under AI replies • suggestion chips row • input bar • per-message "sources" expander listing tool calls
Settings	Grouped list per FR-SET-001 • consents dashboard with status pills • data section with backup status/last run

25.2 Key user flows

- F1 Capture-to-insight: FAB → amount → (suggested category) → Save
→ orchestrator debounce → budget/pace rules → optional insight card
- F2 Receipt: Scan → capture → auto-extract → review/fix → Save
→ duplicate check → transaction + encrypted attachment
- F3 Purchase decision: Advisor → Purchase Advisor → inputs → verdict
→ [Create goal | Remind me | Ignore] → decision trace stored
- F4 Crunch rescue: nightly forecast job finds crunch day (T-9)
→ critical notification → forecast detail → cause list
→ suggested moves (shift payment, pause goal month, cut category)
- F5 Monthly ritual (1st of month): digest notification → review screen:
last month summary → budget suggestions → goal digest → FHS change
- F6 Backup/restore: Settings → enable backup → passphrase + recovery
phrase ack → nightly E2EE blob → new device: restore → decrypt → done

26. Roadmap & Milestones

Phase	Scope	Exit criteria (all MUSTs of listed areas)
Phase 0 — Foundation (4–6 wks)	Module skeleton, design system, encrypted DB, Money/time core, CI, onboarding, accounts, manual transactions	F1 add-txn ≤ 3 taps; airplane-mode pass; migration test harness live
Phase 1 — Core finance (6–8 wks)	Categories + auto-categorisation v1, recurring detection, budgets + alerts, dashboard v1, search/filters, export/import, widget	Classification $\geq 85\%$ on eval set; budget alerts; export round-trip
Phase 2 — Wealth & goals (6–8 wks)	All account types, loans + amortisation, investments + XIRR, gold/crypto valuation via market API, goals engine, net-worth snapshots, backup E2EE	Net worth accurate vs fixtures; goal math golden tests; restore on fresh device
Phase 3 — AI core (8–10 wks)	Fixed/variable engine, forecasting + crunch alerts, emergency-fund coach, FHS, insight orchestrator + feed, notification policy, OCR scanning	Forecast MAPE tracked; OCR total $\geq 95\%$; FHS explainability review
Phase 4 — Advisor (8–10 wks)	Purchase Advisor, simulators, seasonality, vehicle prediction, investment intelligence, chat assistant (on-device LLM), Hindi + 2 languages	PA verdict traces stored; chat guardrail eval green; localisation QA
Phase 5 — Expansion	Household mode, appliances, insurance module, tax planning, business mode, Account Aggregator integration, iOS via KMP	Per-feature PRD addenda

- Release train: internal alpha at end of each phase; public beta after Phase 3; 2-week release cadence thereafter with feature flags.
- Risk register (top 3): (1) categorisation accuracy on Indian merchants → mitigate with KB investment + correction learning; (2) on-device LLM footprint/perf → mitigate with strict tool-call design and dynamic delivery; (3) scope creep → this SRS + ADR discipline is the mitigation.

27. Future Features (design-for, don't build)

Feature	Forward-compatibility decision made now
Insurance module	accounts.type extensible enum; renewal_items table already generic; protection pillar of FHS reads self-declared flags that this module will replace with real policies
Tax planning (India)	categories carry tax_relevance tag field (80C/80D/HRA...) from v1 so history is analysable retroactively
Family/household finance	profile_id on every table from day one; household_members table exists; sharing = future E2EE profile-key exchange
Business mode	tags support business/personal from v1; GST fields captured by OCR are stored even before business reports exist
Account Aggregator (AA)	transactions.source enum includes reserved value 'aa'; import_batches supports statement-grade provenance
iOS / KMP	ARC-002 keeps all engines pure Kotlin; only UI and platform services need porting

28. Requirements Traceability & Change Control

- A traceability matrix (requirement ID → module → tests → screen) is maintained as `/docs/traceability.csv` in the repo and updated in the same PR as the code.
- Changes to any MUST requirement need an ADR referencing the SRS section, the reason, and the impact; the SRS version increments and a change-log entry is added below.
- This document is the single source of truth. If code and SRS disagree, either the code is a bug or an ADR is missing.

Version	Date	Changes
1.0	Jul 2026	Initial master blueprint covering vision, requirements, AI engines, data model, architecture, security, UX, roadmap
1.1	Jul 2026	Added §29 Financial Rulebook & Heuristics KB (RULE-KB) and §30 Market Signal Engine (AI-MKT) with buy-day verdicts, capacity gates, backtesting honesty layer, tranche policy; PERSONAL_MODE flag introduced
1.2	Jul 2026	AI-MKT upgraded: SIG-VALUATION & SIG-VIX added with evidence-weighted scoring, OI/PCR demoted to context tier (MKT-006), three-way lump-sum benchmark (MKT-007), new market data feeds; added §31 Philosophy Charter (Easy-Grow-Control) with Daily Rhythm and §32 Worth-It Engine (AI-WRT)
1.3	Jul 2026	Added §13.3 Buy List & Adaptive Purchase Interview (AI-PA-INT): question ladder scaling with purchase weight vs income, WantScore outcomes (keep/park/suggest-remove), duplicate-ownership & cheaper-alternative finder, buy-timing watch, new wishlist tables and chat tools
1.4	Jul 2026	Growth gap review: added §33 Growth Optimizer Suite (AI-GRW) — Tax Optimizer Lite (80C/80D meter, regime comparator, LTCG lot awareness), Investment Cost Guard (expense-ratio drag), Cash Yield Optimizer (FD laddering, maturity watch), Wealth Trajectory (Future You projection with levers), Windfall Playbook; FHS Growth Efficiency sub-signal
1.5	Jul 2026	Deep review: 6 consistency fixes (cross-refs, tool registry, DB inventory ~75 tables, module list, API endpoints, typo); added §34 Vigilance Pack (anomaly watch AI-ANM, multi-currency/FX consolidation, floating-rate repo-linked loan intelligence, backup restore drill) and §35 Deep Review Record with scope coverage matrix and known limitations
1.6	Jul 2026	Added §8.3 Nature Classification (AI-CLS-N): auto-classification of every transaction into Need/Want/Investment/Asset/Liability with decision order, EMI principal-interest auto-split, True Spend metric separating consumption from conversion, 50/30/20 ring, learning loop; transactions schema gains nature columns
1.7	Jul 2026	Integrated external gap analysis: §36 Financial Order of Operations waterfall (AI-FOO), §37 FI Engine (savings-rate KPI, years-to-FI, India 30–33× corpus, FIRE variants) + net-worth benchmarks, §38 Tax Engine v2 (FY2025-26 parameters, post-Jul-2024 capital gains, harvesting), §39 Protection Suite (insurance adequacy HLV, estate/nomination, household factors), §40 Debt & Credit Discipline (FOIR bands, MAD trap, grace-period, BNPL), §41 Rent-vs-Buy (PTR), §42 Allocation Discipline (glide paths, rebalance-via-inflows, SIP step-up, 110-minus-age), §43 Behaviour & Cost-Drag Dashboard, §44 compliance boundary for future distribution

29. Financial Rulebook & Heuristics Knowledge Base (RULE-KB)

The Rules Engine (L3) ships with a named, transparent rulebook of proven personal-finance heuristics, adapted for India. Every rule is a data row, not hardcoded logic: engines evaluate rules generically, cite them by ID in evidence ("flagged by RULE-EMI-40"), and every threshold is user-editable. Defaults are the floor; personal learned behaviour and user overrides refine them.

29.1 Rule schema

```
rules_knowledge_base(rule_id PK, domain, name, formula_id, params_json,
    severity, rationale, source_note, user_override_json NULL, enabled INT,
    version, created_at, updated_at)
```

```
evaluation: RuleEngine.evaluate(ruleId, FeatureSnapshot) → RuleResult
{ pass|warn|fail, measuredValue, threshold, evidence, ruleId, version }
```

29.2 Saving & budgeting rules

Rule ID	Definition (defaults)	Consumed by
RULE-50-30-20	Needs ≤ 50%, wants ≤ 30%, savings ≥ 20% of income; auto-flexes to observed fixed load (metro preset 60/20/20)	Budget suggester, FHS discipline pillar
RULE-PAY-FIRST	Goal/saving contributions scheduled on salary-credit day, not month-end	Goal engine, notifications
RULE-SAVE-RATE	Savings rate ≥ 20% good, ≥ 30% excellent, < 10% flag	FHS, monthly review
RULE-COOL-OFF	Discretionary purchase > 1% of annual income → Purchase Advisor + optional 24h cooling-off reminder	Purchase Advisor entry
RULE-EMERG-FIRST	No goal below Emergency Fund in waterfall funds while runway < 3 months	Goal waterfall
RULE-LIFESTYLE	If income rises ≥ 10%, suggest ≥ 50% of the raise to savings before budgets inflate	Insight feed

29.3 Investment rules

Rule ID	Definition (defaults)	Consumed by
RULE-AGE-EQUITY	Equity allocation band = (100 – age)% ±10pp (110 – age for Growth profile)	Allocation analysis §11
RULE-5-25	Rebalance when a class drifts 5pp absolute or 25% relative from target	Rebalancing signal
RULE-GOLD-CAP	Gold ≤ 10% of portfolio (flag above)	Diversification
RULE-CRYPTO-CAP	Crypto ≤ 5% of portfolio (flag above)	Diversification
RULE-HORIZON	Money needed < 3y → savings/FD/debt; 3–5y → hybrid; > 5y → equity-eligible	Goal funding buckets
RULE-SIP-STREAK	Missed SIP month → insight with long-term cost of the gap	SIP consistency
RULE-IDLE-CASH	Liquid balance > (STS needs + buffer) for 60 days → deployment prompt	Idle-cash detector, AI-MKT gate
RULE-CONC-15-70	Single holding ≤ 15%, single asset class ≤ 70%	Diversification score

29.4 Debt & big-purchase rules

Rule ID	Definition (defaults)	Consumed by
RULE-EMI-40	Total obligations (EMIs + rent) $\leq$ 40% of income; hard fail at 50%	Purchase Advisor gate 3, FHS debt pillar
RULE-20-4-10	Vehicle: $\geq$ 20% down, loan $\leq$ 4 years, total transport cost $\leq$ 10% of income	Purchase Advisor (vehicle), Vehicle screen
RULE-HOME-EMI	Home: EMI $\leq$ 35% of income; price $\leq$ 5 $\times$ annual income; rent $\leq$ 30%	Purchase Advisor (property)
RULE-CC-UTIL	Credit-card utilisation $\leq$ 30%; never revolve (interest flag = severe)	FHS debt pillar, card alerts
RULE-PAYOFF-ORDER	Debt payoff simulator offers avalanche vs snowball with interest delta computed	Debt simulator
RULE-PREPAY-VS-INVEST	Compare loan rate vs expected return net of tax; show breakeven, let user decide	Loan simulator

29.5 Protection rules

Rule ID	Definition (defaults)	Consumed by
RULE-TERM-10X	Term life cover $\approx$ 10–15 $\times$ annual income if dependents exist	FHS protection pillar
RULE-HEALTH-COVER	Health cover $\geq$ ₹5–10L per family (metro: 10L)	FHS protection pillar
RULE-RUNWAY-M	Emergency runway $\geq$ personal multiplier M (§10)	Emergency coach, AI-MKT gate

RULE / NOTE: Rulebook governance: rules are versioned; changing a threshold records who/when/why in audit_log; engines always cite ruleId + version so past insights stay reproducible (AI-ARC-006).

30. Market Signal Engine (AI-MKT) — Opportunity Detection & Buy-Day Verdicts

Purpose: remove the daily burden of watching markets. The engine continuously monitors user-selected instruments (indices, MF/ETF NAVs, gold), computes objective technical signals, cross-checks the user's own capacity to invest, and issues a direct daily verdict — including "today is a strong day to deploy" — with a full reasoning trace and honest historical statistics.

RULE / NOTE: PERSONAL_MODE: this build is a single-user personal application. A build-config flag PERSONAL_MODE=true enables direct verdict language ("Strong buy day") and removes advisory disclaimers. The transparency rules (P-02, AI-ARC-004) still apply in full: every verdict must be reproducible from data, and predictions are always accompanied by their measured historical hit rate — the app persuades with evidence, never with certainty it does not have.

30.1 Data inputs

- Daily OHLC series for watched instruments via the Market Data API (extends §16.1 with /v1/market/history?ids=...&days=...; 5 years of history cached on first watch, incremental daily updates).
- User capacity snapshot: idle cash beyond buffer (RULE-IDLE-CASH), emergency runway (RULE-RUNWAY-M), 30-day crunch-day forecast (AI-FCT).
- User's signal configuration and tranche policy (§30.4).
- Optional online news context for the LLM explanation layer — clearly labelled "context", never an input to the score.

30.2 Signal library (all objective, all backtestable)

Signal	Definition (defaults, editable)	Score contribution
SIG-VALUATION	Index P/E (and P/B) percentile vs its own 10-year history; bottom quartile scores, bottom decile maxes. Strongest research support (valuation mean-reversion / CAPE literature): low starting valuations → better 3–5y forward returns	0–25 pts
SIG-DRAWDOWN	% below 52-week high; tiers at –5 / –8 / –12 / –20%	0–20 pts
SIG-VIX	India VIX percentile vs 5y history; top quartile scores, top decile (panic/capitulation zone) maxes — implied-vol spikes historically coincide with above-average forward returns	0–15 pts
SIG-MA200	% deviation below 200-day moving average; tiers at –2 / –5 / –10%	0–15 pts
SIG-RSI	RSI(14) < 30 oversold (=10 pts), < 25 deeply oversold (=15)	0–15 pts
SIG-RARITY	Percentile of today's price vs trailing 12 months (bottom decile = max)	0–5 pts
SIG-STREAK	N consecutive down days (≥ 4 = 3 pts, ≥ 6 = 5)	0–5 pts

- Context-tier displays (shown, never scored):** SIG-SUPPORT (proximity to recent held lows — descriptive, levels shift), SIG-YIELD-GAP (index earnings yield vs 10-yr G-Sec — contested "Fed model" evidence, useful colour only), and OI/PCR per MKT-006 below.

- **Signal quality note baked into the UI:** valuation + fear (VIX) carry the highest weights because they have the strongest published evidence for medium-term forward returns; pure price-pattern signals are capped low. The rule editor shows each signal's own backtest so weights can be personalised on evidence.

OpportunityScore(instrument, day) = Σ signal points // 0–100+, capped 100

CapacityGate (must ALL pass for a deploy verdict):

runwayMonths $\geq$ M (RULE-RUNWAY-M)
 idleCash $>$ 0 (RULE-IDLE-CASH beyond buffer)
 no crunch day in 30-day forecast including the proposed deployment

Verdict:

score $\geq$ 70 and gate pass $\rightarrow$ STRONG BUY DAY (rare by construction)
 50–69 and gate pass $\rightarrow$ GOOD BUY DAY
 30–49 $\rightarrow$ NEUTRAL – SIP as planned
 gate fail (any score) $\rightarrow$ NOT YOU, NOT NOW (capacity reason shown)
 score $<$ 30 $\rightarrow$ NO EDGE TODAY

30.3 Honesty layer — backtesting (the core differentiator)

ID	Requirement
MKT-001	Every signal and every composite threshold MUST be backtested on the instrument's cached 5-year history: number of trigger days, forward 30/90/365-day return distribution (median, P25–P75, % positive), vs. all-days baseline.
MKT-002	Every verdict card MUST display its backtest line, e.g. "Score $\geq$ 70 occurred 11 times in 5y; 90 days later Nifty was higher 9/11 times, median +4.2% (all-days baseline: +2.1%)."
MKT-003	The engine MUST track its own live verdicts (verdict, price, date) and show a running scoreboard (paper performance), exactly as forecast accuracy is tracked (AI-FCT-004). If a signal's live hit rate degrades below baseline, the app says so.
MKT-004	The LLM MAY add news context ("Markets weak amid Middle-East tensions") when online, always labelled as context; the score and verdict MUST be computable and identical with news unavailable (P-04).
MKT-005	No signal may use unverifiable inputs (analyst targets, sentiment scores from third parties) — only price/volume/valuation series and the user's own finances.
MKT-006	Options-market data (Put-Call Ratio, Open Interest walls, Max Pain) is CONTEXT-TIER ONLY: it may be displayed on the verdict card labelled "positioning context — no proven multi-week edge; excluded from score", and MUST NOT contribute score points. Rationale recorded in-app: PCR shows mixed, days-horizon evidence at best; Max Pain lacks rigorous support; OI walls describe positioning that rolls daily — all mismatched to a multi-year accumulation horizon. Promotion to a scored signal is allowed only if the user's own backtest panel demonstrates edge above baseline on that instrument.
MKT-007	Three-way benchmark: SIG-REVIEW MUST compare (a) the tranche strategy vs (b) plain monthly SIP vs (c) lump-sum on day one, over the trailing 1y/3y — because research (Vanguard-style immediate-vs-averaging studies) shows lump-sum wins $\sim$ 2/3 of the time; dip-buying must prove it beats that bar, not zero.

30.4 Deployment policy — tranches, not all-in

- User defines a tranche policy per instrument: e.g. deploy 25% of idle cash at GOOD, 50% at STRONG, keep $\geq$ 25% dry powder; minimum 7 days between tranches (prevents catching every day of a long slide).
- Verdict card computes the exact suggested amount from the policy and current idle cash, and shows the after-deployment forecast (no new crunch days — re-verified).

- One-tap actions: "Log investment" (creates the transaction + holding lot), "Snooze 3 days", "Adjust my rules".
- A monthly SIG-REVIEW digest compares tranche deployments vs plain monthly SIP for the trailing year — so the strategy continuously proves (or disproves) itself.

30.5 UX — the Opportunity screen

Element	Blueprint
Watchlist header	Instruments with today's score gauge, verdict chip, and price vs 52w-high sparkline
Verdict card	Verdict banner • signal table (each signal: value, threshold, points) • capacity gate rows (pass/fail with amounts) • backtest line (MKT-002) • suggested tranche amount • actions
History tab	Past verdicts scoreboard: date, verdict, price then, price now, outcome vs baseline (MKT-003)
Rule editor	Sliders per signal threshold with instant backtest re-run ("with this setting: 23 triggers in 5y, 74% positive at 90d")
Notification	Fires only on GOOD/STRONG with gate pass, max 1/day/instrument, critical-tier optional (\$17 policy applies)

30.6 Engine placement & module

- New module :domain:engines:marketsignal (pure Kotlin, ARC-002 compliant); indicator math (MA, RSI, drawdown, percentile) implemented with golden-file tests against known series.
- New tables: watchlist, price_series (instrument_id, date, o/h/l/c, volume), signal_configs, signal_verdicts (full trace persisted, AI-ARC-006), backtest_results (cached, invalidated on config change).
- Chat tools added to §19.2 registry: market_status(instrument), opportunity_check(instrument), backtest(signal_config) — so you can ask "is today a good day to buy Nifty?" and get the same evidence-traced verdict in chat.
- Roadmap: enters Phase 4 alongside the Purchase Advisor (shared verdict-card UI components); price-history API lands with Phase 2 market data work.

31. Product Philosophy Charter — Easy · Grow · Control

This section states the app's soul in one page, so every future feature can be tested against it. The app exists to do three things for its user, every single day:

Pillar	Promise	What it means in features
EASY	Money management costs the user minutes, not hours. The app does the searching, watching, remembering, and math.	Auto-capture (OCR/SMS/recurring), auto-categorisation, market watching (AI-MKT), renewal & bill memory, one daily brief instead of ten apps
GROW	Every idle rupee gets noticed and offered a job. Wealth compounds by default, not by willpower.	Idle-cash detection, opportunity verdicts with backtested evidence, goal waterfall, SIP consistency, prepay-vs-invest math, lifestyle-inflation guard
CONTROL	The user is always the CFO's boss: every number explainable, every rule editable, every action user-executed, all data user-owned.	Evidence chips on every claim, rulebook overrides, no auto-execution, full export, erase-all, consents dashboard

RULE / NOTE: Feature admission test: a proposed feature must strengthen at least one pillar and weaken none. A feature that grows money but removes control (auto-trading), or adds ease but hides reasoning (black-box scores), is rejected or redesigned.

31.1 The Daily Rhythm — how the app shapes a day

Moment	Behaviour
Morning brief (one notification, ~8:00, configurable)	Safe-to-Spend today • anything due in 48h • market verdict for watched instruments (only if GOOD/STRONG or gate-relevant) • one insight max. Everything else waits.
During the day	Silent capture: transactions land via SMS/quick-add; Purchase Advisor is one tap away when standing in a shop; no mid-day nags.
Evening (optional, off by default)	Capture nudge only if the day has zero transactions and history says that's unusual.
Weekly (Sunday evening)	Digest: week's spend vs pace, insight roll-up, uncategorised backlog, upcoming week's obligations.
Monthly (1st, salary-aware)	The ritual (flow F5): last month verdict, budget suggestions, goal digest, FHS change, Worth-It review (§32), SIG-REVIEW benchmark.

- Rhythm rules: the app initiates contact at most once per day outside critical events (§17 policy is the enforcement); every touchpoint must be completable in under 60 seconds; the monthly ritual is the only intentionally long session (~10 min).
- Metric of success for this section: total weekly time-in-app trends DOWN after month 3 while coverage and follow-rate stay high — the CFO is working so the user doesn't have to.

32. Worth-It Engine (AI-WRT) — Was Buying That Worth It?

The Purchase Advisor (§13) answers "can I afford it?" before buying. The Worth-It Engine answers the other half of your philosophy — **"was it worth it, and what does that teach the app about me?"** — turning past purchases into a personal value model that makes future advice smarter.

32.1 Capabilities

Capability	Definition
Cost-per-use tracking	For durables (gadgets, vehicles, appliances, gear) the user can log or estimate usage; the app computes ₹/use and ₹/month of ownership. "The ₹24,000 headphones: ₹32/use so far — trending worth it. The ₹18,000 treadmill: ₹1,500/use — talk to me."
Subscription utilisation	Recurring subscriptions get a lightweight usage check-in quarterly (used a lot / sometimes / barely); barely-used subs are flagged with annualised cost and a cancel reminder. Price-hike detection compares renewal amounts.
30-day regret check-in	For purchases above the RULE-COOL-OFF threshold, one question 30 days later: "Worth it?" (Yes / Mixed / Regret) + optional reason chip. Two taps, skippable, never nags twice.
Personal value model	Regret ratings + cost-per-use aggregate into per-category value scores: "Your travel spends: 92% rated worth it. Gadget spends: 41%." This is the user's own revealed preference — no external judgement.
Feedback into Purchase Advisor	A new PA gate 8 — VALUE HISTORY: "Similar purchases (electronics > ₹15k): you rated 3 of 5 as regret; average usage fell 70% after month 2." Advisory weight only; never a hard fail (the user's taste may change — CONTROL pillar).
Time-cost translation	Optional display: price expressed in hours of the user's own work (from income), e.g. "₹6,000 ≈ 1.4 working days." Well-studied framing that measurably reduces impulse purchases; toggleable because some users find it heavy.
Lifestyle-inflation guard	Tracks category spend as % of income across income changes; if income +15% and dining +45%, a gentle annual insight quantifies the compounding cost of the drift (extends RULE-LIFESTYLE).

32.2 Rules

ID	Rule
WRT-001	All Worth-It data is self-reported and private; it never leaves the device, is excluded from cloud LLM context packs, and is deletable per record.
WRT-002	Tone is non-judgemental by design: the engine reports the user's own past ratings back to them; it never labels a category "waste". Copy review against §24.1 tone rules is mandatory.
WRT-003	Check-ins obey the notification budget (§17); a skipped check-in is never re-asked for that purchase.
WRT-004	Value-model scores require ≥ 5 rated purchases in a category before they influence Purchase Advisor; below that they render as "learning".
WRT-005	New tables: usage_logs(item/asset, date, note), purchase_reviews(txn_id, rating, reason, asked_at), value_model_scores(category, score, n, updated_at). PA decision traces record whether gate 8 fired (AI-ARC-006).

- Placement: :domain:engines:worthit module; check-ins ride the existing notification engine; monthly ritual gets a "Worth-It minute" card. Roadmap: Phase 4, after Purchase Advisor lands (it feeds on PA's cool-off threshold and decision traces).

33. Growth Optimizer Suite (AI-GRW) — Closing the Silent Growth Leaks

A gap review of v1.3 against the full money-growth loop (earn → keep → save → deploy → compound → protect from drag) found the app strong on keeping, saving, and deploying, but missing four **silent drags on compounding**: tax, fees, lazy cash yield, and unplanned windfalls — plus the motivational anchor of seeing the long-term trajectory. This suite closes them. All modules obey the charter (§31): compute and suggest, never execute; every number evidence-linked.

33.1 Tax Optimizer Lite (pulled forward from Phase 5)

ID	Requirement
GRW-TAX-001	Section 80C / 80D utilisation meter: transactions tagged tax_relevance (already captured since v1, §27) roll up into a live "₹87,000 of ₹1.5L 80C used" meter; from December onward a monthly insight shows the remaining gap and the marginal tax saved by filling it (at the user's slab).
GRW-TAX-002	Old vs New regime comparator: given salary structure inputs (basic, HRA, rent paid) and tagged deductions, compute tax under both regimes and show the annual difference. Assumption set is versioned per financial year and user-editable; the app states its FY rules version on the result.
GRW-TAX-003	Capital-gains awareness: for equity holdings, track LTCG/STCG status per lot (holding_lots already store dates); alert when (a) the ₹1.25L annual LTCG exemption is unused in Q4 — harvesting opportunity, and (b) a planned sale is within 30 days of turning long-term ("waiting 24 days changes the tax rate on this gain").
GRW-TAX-004	Scope guard: the module computes and educates from user data; it never files, never interprets edge cases, and links out to a professional for complex situations. Tax rule tables ship as versioned knowledge base updated yearly via /v1/knowledge/tax.

33.2 Investment Cost Guard — fees are negative compounding

ID	Requirement
GRW-FEE-001	The holdings table gains expense_ratio_bps and plan_type (direct/regular) fields; MF holdings auto-fill expense ratio from the NAV feed metadata where available, else manual.
GRW-FEE-002	Fee-drag calculator on every holding and the portfolio: projected cost of current expense ratios over 10/20 years vs a direct/low-cost equivalent, in rupees ("this regular plan's extra ~1% ≈ ₹4.8L over 20 years on your SIP"). Regular-plan holdings get a one-time educational insight about direct plans.
GRW-FEE-003	Portfolio weighted-average expense ratio is a tracked metric on the Investments screen; rising WAER triggers a review insight.

33.3 Cash Yield Optimizer — no rupee sleeps at 3%

ID	Requirement
GRW-YLD-001	Yield map: every liquid/near-liquid account carries its interest rate (manual or preset); a "cash yield" card shows blended yield on total cash vs the current FD/liquid-fund rate band (feed already in §16.1) and the annual rupee cost of the gap.
GRW-YLD-002	FD laddering assistant: for the emergency fund above the instant-access floor (default 1 month of essentials stays instant), propose a 3/6/9/12-month FD ladder sized so one rung matures every quarter — liquidity preserved, yield raised; show the ladder as editable before the user acts at their bank.
GRW-YLD-003	Maturity watch: FD/RD maturity dates tracked; alert 7 days before with reinvest-vs-redeploy options (auto-renewal at poor rates is a known leak).

33.4 Wealth Trajectory — "Future You" screen

ID	Requirement
GRW-TRJ-001	Project net worth 5/10/20 years out from: current net worth, trailing-12-month savings rate, current allocation's expected return (user assumptions from §15), inflation (CPI feed) — shown in both nominal and today's-rupees terms, with P25–P75 bands via seeded Monte Carlo.
GRW-TRJ-002	Interactive levers with instant recompute: savings rate $\pm$, allocation shift, fee reduction (from §33.2), one-time windfall — each lever shows its 20-year delta so the user sees which knob actually matters (" +5% savings rate: +₹38L; picking better dip days: +₹2.1L" — an honest ranking of effort vs payoff).
GRW-TRJ-003	Trajectory recomputes monthly and the ritual (F5) shows drift vs last year's projection — the compounding scoreboard.

33.5 Windfall Playbook

ID	Requirement
GRW-WND-001	Windfall detection: a credit $> 2\times$ median monthly income (bonus, gift, maturity, refund) triggers the playbook — a suggested split computed from current state: top-up emergency gap $\rightarrow$ kill highest-rate debt above threshold $\rightarrow$ tax-saver gap (§33.1) $\rightarrow$ goal waterfall / opportunity deployment (AI-MKT gate-aware) $\rightarrow$ an explicit guilt-free enjoyment slice (default 10%, editable — sustainable plans budget joy).
GRW-WND-002	The split is a draggable plan the user adjusts and confirms; confirmed slices become linked goal contributions/transactions so follow-through is tracked, not assumed.

- Placement: :domain:engines:growth module; tax KB via new /v1/knowledge/tax endpoint; new fields on holdings and accounts (expense_ratio_bps, plan_type, interest_rate_bps); Wealth Trajectory joins the Advisor hub. Roadmap: GRW-YLD and GRW-TRJ land in Phase 3 (they need only existing data), GRW-TAX and GRW-FEE in Phase 4, GRW-WND with the goal waterfall in Phase 2.
- FHS hook: a small "Growth Efficiency" sub-signal (fee WAER, cash-yield gap, tax-gap in Q4) feeds the Protection & Growth pillar so the score rewards fixing silent leaks.

34. Vigilance & Completeness Pack

Additions from the v1.4 deep review: features that protect money already earned and remove real-world blind spots the earlier sections assumed away.

34.1 Transaction Anomaly Watch (AI-ANM)

ID	Requirement
ANM-001	Duplicate-charge detection: same merchant + same amount within 48h (and not a known recurring cadence) → "Charged twice by X?" insight with both transactions side by side and a dispute checklist. Runs on every commit.
ANM-002	Unusual-amount detection: a transaction > 3× the user's median for that merchant (min 5 prior samples) or > P95 of the category → gentle verification insight, batched, never a scary alert.
ANM-003	Subscription creep: renewal amount > last cycle → price-hike insight with annualised delta (consolidates the §32 capability under this engine); new small recurring charges the user never confirmed → "Is this yours?" review.
ANM-004	Dormant-account drain: outflow from an account with no activity for 90+ days → verification insight.
ANM-005	All anomaly insights are advisory verifications (the app cannot see bank truth); tone per WRT-002; false-positive feedback ("this is fine") suppresses the pattern (fingerprint suppression, §7.2).

34.2 Multi-Currency & FX Consolidation

ID	Requirement
FX-001	Accounts, holdings, and transactions already carry a currency code; net worth, forecasts, and the FHS MUST consolidate to the profile base currency (default INR) using daily cached FX reference rates (/v1/market/fx). Non-base assets show native + converted amounts with the rate date.
FX-002	US stocks / RSU / ESPP holdings (common for the primary persona) are first-class: USD holding valued via price feed × FX; unrealised FX gain/loss is displayed separately from asset gain so the user sees what the rupee did vs what the stock did.
FX-003	Offline: last cached rate is used with a staleness label (EXT-002 applies); FX rates never block any computation.
FX-004	Cross-currency transfers record both amounts + the effective personal rate; fees implied by rate spread are surfaced ("this remittance cost ~₹412 vs reference rate").
FX-005	New table fx_rates(pair, date, rate, source); Money stays integer minor units per currency — conversion happens at read/display and analytics layers, never by mutating stored amounts.

34.3 Floating-Rate Loan Intelligence (repo-linked EMIs)

ID	Requirement
FLT-001	loan_details gains: rate_type (fixed/floating), benchmark (repo/MCLR/other), spread_bps, reset_frequency. Most Indian home loans are repo-linked; the RBI repo feed (§16.1) finally has a consumer.
FLT-002	On a repo change, recompute the projected EMI or tenure impact for each floating loan and notify: "Repo -25 bps: your home loan EMI should drop ~₹690/month (or tenure -7 months). If your bank hasn't passed it on by the next reset, ask."
FLT-003	Forecast integration: floating EMIs in the 12-month forecast carry rate-scenario bands (±50 bps) so long forecasts are honest about rate risk.
FLT-004	The prepay-vs-invest simulator (RULE-PREPAY-VS-INVEST) MUST use the loan's current effective rate, not the origination rate.

34.4 Backup Restore Drill

ID	Requirement
DRL-001	Twice a year, prompt a 60-second restore drill: decrypt the latest backup in a sandbox, verify integrity (row counts, checksums), and confirm the passphrase still works — because an untested backup is a hope, not a plan. Result logged in <code>backups_log</code> ; failures raise a critical insight.

- Placement: `:domain:engines:anomaly` module (ANM); FX in `:core:model` + repository layer; FLT extends the loan engine; DRL in `:sync:backup`. Roadmap: FX-001..005 and FLT in Phase 2 (they touch core money math and loans — cheaper early), ANM in Phase 3 with the orchestrator, DRL with backup in Phase 2.

35. Deep Review Record & Scope Coverage Matrix (v1.5)

35.1 Consistency review — issues found and fixed in this version

#	Issue	Resolution
1	Stale cross-reference: §2.3 pointed Account Aggregator readers to §17 (now Notifications)	Corrected to §27
2	Chat tool registry (§19.2) missing tools introduced by §13.3 and §30	Registry consolidated; convert_currency added
3	DB inventory (§20.1) missing 12 tables introduced in v1.1–v1.4	New "Rulebook, signals & advisor" domain row; total revised to ~75
4	Module list (§21.2) missing market signal, worthit, growth engines	Added, plus new anomaly module
5	API table (§22.2) missing /v1/market/history and /v1/knowledge/tax referenced elsewhere	Both added, plus /v1/market/fx
6	Typo in GRW-FEE-001	Fixed
7	No multi-currency/FX consolidation despite currency codes on every money row	§34.2 added
8	Repo-rate feed existed with no consumer for floating-rate EMLs	§34.3 added

35.2 Scope coverage matrix — original request vs document

Originally requested item	Covered	Where
Executive summary and vision	Yes	§1
Business goals	Yes	§2
Target users and personas	Yes	§3
Functional requirements for every screen	Yes	§5 (10 screen groups, 60+ FRs)
Detailed AI architecture	Yes	§7 (6-layer pipeline + orchestrator)
Database schema (100+ tables if needed)	Yes	§20 (~75 normalised; ceiling reserved)
System architecture diagrams	Yes	§7, §21.1 (text diagrams; hi-fi visuals are a design-phase task)
Android app architecture (Compose, Room, Repository, ViewModel...)	Yes	§21 (modules, stack, rules, testing)
AI decision engine design	Yes	§13 (+ §13.3 interview)
Fixed vs variable expense prediction	Yes	§8
Emergency fund calculation logic	Yes	§10
Investment intelligence module	Yes	§11 (+ §30, §33.2)
Vehicle maintenance prediction module	Yes	§12
Purchase advisor decision engine	Yes	§13
Financial health scoring methodology	Yes	§14 (+ Growth Efficiency sub-signal)
Goal planning engine	Yes	§15
Cash-flow forecasting engine	Yes	§9 (+ FLT-003 rate bands)
External data (inflation, RBI, fuel, gold, market)	Yes	§16 (12 feeds)
Notification engine	Yes	§17
OCR and receipt scanning workflow	Yes	§18 (+ §5.4)
AI chat assistant design	Yes	§19
Privacy and security model	Yes	§23

Originally requested item	Covered	Where
API design	Yes	§22
UI/UX guidelines	Yes	§24
Screen-by-screen wireframes	Yes	§25.1 (textual blueprints)
User flows	Yes	§25.2
Roadmap and milestones	Yes	§26
Future features (insurance, tax, family, business)	Yes	§27 (+ tax pulled forward §33.1)
Rules for writing code & deciding things	Yes	§4, §21.3–21.6, §28, rulebook §29
Grow-money philosophy (Easy·Grow·Control, worth-it, buy-day verdicts)	Yes	§29–§33 (added during this project)

35.3 Known limitations (accepted, on record)

- Wireframes are textual blueprints, not visual mocks — a Figma pass is the first design-phase task.
- AI-MKT signal weights are evidence-informed defaults, not personal guarantees; the backtest panel and MKT-007 benchmark exist precisely because dip strategies often underperform lump-sum.
- Tax Optimizer Lite depends on a yearly-maintained rule KB; complex cases (capital gains on property, foreign assets/RSU taxation) stay out of scope and are flagged to a professional.
- No automatic bank aggregation in v1: coverage depends on SMS parsing + OCR + manual entry; the AA integration path is reserved (§27).
- Single-device single-user by design until Phase 5; the E2EE backup is the only cross-device bridge.

36. Financial Order of Operations — Priority Waterfall (AI-FOO)

The connective tissue the earlier engines lacked: a deterministic answer to "**what should my next rupee do?**". AI-FOO consumes the forecast surplus and outputs a ranked action list with rupee amounts. It supersedes the simple waterfall in §15 and operationalises RULE-EMERG-FIRST as a stage gate. India-adapted sequence:

Stage 0	STARTER BUFFER	min(₹50,000, 1 month essentials) – instant access
Stage 1	CAPTURE EPF/VPF	employer EPF is a guaranteed-match analogue; verify contribution captured; VPF headroom shown
Stage 2	KILL FIRE DEBT	any APR ≥ 13–14% (credit cards 36–42%, costly personal loans): paying a 42% card = guaranteed 42% return – nothing else competes
Stage 3	FULL EMERGENCY	to target M months (§10; stability-scaled)
Stage 4	TAX-ADVANTAGED	80C headroom (EPF/PPF/ELSS), NPS 80CCD(1B) ₹50k – only when the regime comparator (§38) says old regime wins for this user
Stage 5	GOAL INVESTING	>5-year goals to equity-eligible buckets (RULE-HORIZON); shorter goals to debt/FD
Stage 6	GREY-ZONE DEBT	APR ~10–12%: aggressive payoff generally beats expected equity (~12% nominal) after tax & risk – user chooses with the math shown
Stage 7	LOW-RATE DEBT	home loan prepay vs invest → existing simulator (RULE-PREPAY-VS-INVEST, current effective rate)

output: ranked "next best rupee" cards, each with amount, stage, evidence, and one-tap goal/contribution creation; recomputed monthly and on any surplus change; every skipped stage shows why it was skipped

ID	Rule
FOO-001	The waterfall is strict-order by default; user reordering is allowed (Control pillar) but shows the cost of deviation ("funding Goa before the card costs ~₹2,900/quarter in interest").
FOO-002	The Home dashboard and monthly ritual MUST surface the single top waterfall action; the Advisor hub shows the full ranked list.
FOO-003	Stage thresholds are rulebook rows (§29 schema) — editable, versioned, cited.

37. Financial Independence Engine (AI-FI) & Net-Worth Benchmarks

37.1 Savings rate → years to FI

Savings rate — not returns — dominates early accumulation, and it becomes a first-class KPI on the dashboard and in the ritual.

```
savingsRate = (income - trueSpend) / income // §8.3 metric
yearsToFI solved from: corpus(t) reaches target, assuming real return
~5–6% and contributions = savingsRate × income
reference curve (≈5% real, from zero):
10% → ~51y 25% → ~32y 50% → ~17y 65% → ~10.5y 75% → ~7y

corpusTarget = annualExpenses × multiple
baseline 25× (4% SWR) shown for comparison
INDIA DEFAULT 30–33× (SWR 3.0–3.3%) – higher inflation (~6–7%),
no social-security floor; early retirement (40s) → 33×+
+ separate medical buffer ₹25–40L (healthcare inflation ~14%),
excluded from the SWR corpus
This supersedes the §15 default (SWR 4%) – §15 keeps the formula,
defaults now come from this section's India-adjusted table.
```

- FIRE variants selectable as retirement-goal presets: Lean / Standard / Fat / Coast / Barista, each just changing target expenses and contribution rules.
- Assumptions panel is always visible (nominal equity ~12%, inflation ~6–7%, real ~5–6%) with a projection haircut toggle; all figures labelled estimates, never promises.

37.2 Net-worth benchmark overlays

Benchmark	Encoding
Expected net worth	age × pre-tax annual income ÷ 10; ≥ 2× = prodigious accumulator, < 0.5× = under-accumulator — shown as context bands on the net-worth chart
Salary multiples	1× salary by 30, 3× by 40, 6× by 50, 8–10× by 60 — India-softened for later career starts; EPF/PPF/gold included in the numerator
Honesty notes	Compare to medians, not averages (averages are skewed by the ultra-wealthy); benchmarks are context, never judgement — tone per WRT-002

38. Tax Engine v2 (AI-TAX) — supersedes §33.1

38.1 FY2025-26 parameter table (versioned KB, updated every Budget)

Parameter	Value (FY2025-26)
80C cap (old regime only)	₹1.5L across EPF/PPF/ELSS/life premium/tax-saver FD/home-loan principal
80CCD(1B) NPS (old regime only)	Additional ₹50,000 → ₹2L combined with 80C
80CCD(2) employer NPS	Deductible in BOTH regimes, up to 14% of salary
80D health premium	Old regime only
New regime	Standard deduction ₹75,000; 87A rebate → income ≤ ₹12L effectively tax-free; 30% slab starts above ₹24L; no 80C/80D/80CCD(1)/(1B)
Break-even engine	Compute both regimes from the user's actual tagged deductions + salary structure (basic/HRA/rent); old wins only when utilised deductions exceed the computed break-even — shown in rupees

38.2 Capital-gains ledger (post-23-July-2024 rules)

Asset	Treatment
Equity / equity MF	STCG (≤12 mo) 20%; LTCG (>12 mo) 12.5% on gains above ₹1.25L/year, no indexation
Debt MF bought on/after 1 Apr 2023	Slab rate always — no LTCG benefit; the engine labels these lots accordingly
Pre-Apr-2023 debt MF, gold	12.5% LTCG without indexation
Property	12.5% no-indexation, or 20% with indexation if acquired before 23 Jul 2024 and lower — engine computes both
SGB held to maturity	Capital gains fully exempt — flagged as the most tax-efficient gold route in gold insights

ID	Rule
TAX-001	Harvesting alerts (generic, never "sell fund X"): (a) unused ₹1.25L LTCG exemption approaching FY-end — gains can be realised tax-free and cost basis reset; (b) harvestable losses before 31 March (India has no wash-sale rule); (c) STCG→LTCG countdown when a planned sale is < 30 days from turning long-term.

ID	Rule
TAX-002	Every figure comes from the versioned tax KB (/v1/knowledge/tax); the FY rules version is printed on every result; the engine never files and links complex cases (property, foreign/RSU taxation) to a professional.

39. Protection Suite — Insurance Adequacy, Estate & Household (AI-INS)

39.1 Insurance adequacy (replaces the self-declared flags in §14/§11)

```
termCoverNeeded = max(
  10-15 × annual income + outstanding liabilities,
  HLV by IRDAI age multiplier: 25× (18-35), 20× (36-45),
                                15× (45-50), 10× (51-60) )
coverGap = termCoverNeeded - existing term cover // the headline number

healthCoverFloor = ₹10L base per metro family + super top-up,
  sized against ~14% healthcare inflation
```

ID	Rule
INS-001	Cover-gap calculator with policy records (insurer, cover, premium, renewal) feeding renewals (§17) and the FHS Protection pillar with real numbers instead of self-declared booleans.
INS-002	Endowment/ULIP detector: insurance-cum-investment premia are flagged with the buy-term-invest-the-rest comparison — a ₹1Cr term plan at roughly ₹8–12k/yr vs ₹60k–1L/yr for equivalent-cover endowment; the difference SIP-ed at ~12% compounds to ₹1.5–2.7Cr over 30 years while endowments have historically returned ~4–6% — shown as math, decision stays with the user.

39.2 Estate & nomination guard

- Every account/holding gains nomination_status (yes/no/unknown) and holding_mode (single / either-or-survivor / joint); a completeness checklist scores the estate readiness and nags — gently, quarterly at most — on gaps.
- Will awareness: an educational nudge with an India-specific explainer (intestate succession is slow and contested; a simple registered will is high value); the app stores only a "will exists / updated on" flag, never the document by default.

39.3 Household factors (extends §10, §15)

- Single-income households: emergency multiplier +1 (already in §10) AND term-cover multiplier nudged to the top of the 10–15× band.
- Dependent-parents support: first-class recurring obligation type with a contingency line in the forecast — a distinctively Indian planning factor the generic model missed.
- Education goals default to 10–12% inflation, not general CPI — child-education costs outrun CPI; the goal engine (§15) reads this override from the KB.

40. Debt & Credit Discipline Engine (AI-CRD)

40.1 FOIR — the native Indian obligation metric (refines RULE-EMI-40)

FOIR = (all EMIs + proposed EMI) / GROSS monthly income
 bands: green < 40% | caution 40–50% | risk > 50%
 (banks approve up to ~40–55%; the app's planning ceiling stays tighter: total EMI ≤ 40% of TAKE-HOME, home EMI ≤ 30–35%)
 LTV context for home loans: 90% ≤ ₹30L | 80% ₹30–75L | 75% > ₹75L
 Context insight: national EMI-to-income has drifted ~46% (2020) → ~61% (2024); price-to-income ~7.5 (MMR ~14×) – "within bank limits" is not the same as "affordable"; the app plans to the tighter line.

40.2 Credit-trap simulators

ID	Rule
CRD-001	Minimum-Amount-Due trap simulator: at 5% MAD and ~3.5%/month, ₹50,000 balloons to ~₹1.28L over ~11 years (2.5×) — shown interactively with the user's real statement figures whenever revolving is detected; revolving interest is a severe FHS event (RULE-CC-UTIL).
CRD-002	Grace-period optimiser: purchases just after statement date get up to ~48–50 interest-free days vs ~19 just before — the app knows statement days (FR-ACC-002) and suggests timing for large planned purchases (Buy List integration §13.3).
CRD-003	BNPL / "no-cost" EMI true-cost flag: processing fees, forfeited discounts, and GST on interest are computed into an effective APR; a missed instalment reverting to ~36–42% APR is warned upfront.
CRD-004	Utilisation guidance refined: < 30% good, < 10% optimal; per-card and aggregate tracked.
CRD-005	Payoff strategy (extends RULE-PAYOFF-ORDER): avalanche default with snowball option; cutoffs from §36 — APR ≥ 13–14% is "financial house on fire" priority, ~10–12% is the grey zone where payoff generally beats investing; both simulations show total-interest and debt-free-date deltas.

41. Rent-vs-Buy Engine (AI-RVB)

PTR = property price / annual rent // rental yield = 1 / PTR
 PTR < 20 → buying competitive (yield > 5%)
 20–25 → neutral — decide on non-financial factors
 PTR > 25 → renting + investing the difference usually builds more wealth (Indian metros commonly run 30–45×)

full comparison (15–20y, side by side):

BUY: down payment + stamp duty (4–7%) + interest + maintenance
 – 24(b)/80C tax benefits + property appreciation (assumption)
 RENT: rent (escalating) + SIP of (EMI – rent) + down-payment corpus
 invested at the user's equity assumption
 output: net-worth curves for both paths, crossover year, and the
 assumption sliders (appreciation, rent escalation, returns)

- Integrates with the Purchase Advisor for home purchases (RULE-HOME-EMI, FOIR §40.1) and uses the effective loan rate from §34.3.
- Verdict language stays comparative ("under these assumptions, renting + investing ends ~₹38L ahead") — assumptions are the user's, clearly visible.

42. Allocation Discipline — Glide Paths, Rebalancing, SIP Step-Up (extends §11, §29.3)

ID	Rule
ALC-001	Default equity starting band moves to 110 – age (RULE-AGE-EQUITY updated); the 100-minus-age variant remains selectable but is labelled conservative for long horizons. Goal-based glide paths always override the age rule.
ALC-002	Glide-path scheduler: every dated goal gets an automatic de-risking plan — equity exposure for that goal's corpus steps down via STP-style shifts over the final 3 years (RULE-HORIZON enforced), with reminders at each step.
ALC-003	Rebalance-via-inflows preference: when drift exceeds the ± 5 pp band (RULE-5-25), the engine FIRST proposes directing new SIP inflows to the underweight class — rebalancing without selling avoids realising LTCG; sell-based rebalancing is the fallback with its tax cost computed (§38).
ALC-004	SIP step-up: default nudge of +10% annually aligned to salary-revision month; the Wealth Trajectory lever (§33.4) shows its 20-year delta so the nudge sells itself.

43. Behaviour & Cost-Drag Dashboard (AI-BEH)

ID	Rule
BEH-001	Behaviour-gap monitor: investor returns lag fund returns mainly through timing churn (US evidence: ~ 1.2 pp/year lost; directionally global, exact figure US-specific — labelled as such). The engine tracks the user's own churn (redemption/switch frequency, SIP pauses in drawdowns) and computes their personal gap: money-weighted (XIRR) vs time-weighted holding return per holding.
BEH-002	Direct-vs-regular drag: the §33.2 fee guard adds the headline framing — a $\sim 1\%$ TER gap on a ₹10–20k monthly SIP compounds to roughly ₹13.5–19L less over 20 years; shown once per regular-plan holding, then available in the dashboard.
BEH-003	Lifestyle-inflation ratio (refines RULE-LIFESTYLE): trailing-12m spend growth $\div$ income growth; ratio > 0.5 (more than half of raises consumed) $\rightarrow$ annual insight with the compounding cost and the "save $\geq 50\%$ of every raise" nudge.
BEH-004	Money-leaks detector: ATM fees, forex markup / DCC charges (always-decline-DCC education), card convenience fees, idle subscriptions (§34.1) — aggregated into an annualised "leak total" card; individually small, collectively motivating.
BEH-005	Drawdown discipline: during market drawdowns the app's only automated behaviour is to defend the plan — "your SIP bought 12% more units this month" framing; it never suggests pausing SIPs, and pausing triggers a consequence preview.

44. Education-vs-Advice Compliance Boundary (for any future distribution)

This build is PERSONAL_MODE (single user, own device, own data — §30), so direct verdict language stays. This section exists so that any future decision to distribute the app starts from a documented boundary rather than a retrofit.

- The regulated line (India): without SEBI RIA registration, no specific-security buy/sell/hold recommendations, no return/performance claims, and no predictions framed on a named security's recent price data. Generic calculators, asset-class allocation, tax math, and education sit outside that line; advice on gold, real estate, and crypto is outside SEBI's securities purview (with no-SEBI-recourse disclosure).
- By design, every engine in this SRS already outputs at asset-class or user's-own-data level — the only PERSONAL_MODE-dependent behaviours are verdict phrasing (AI-MKT §30) and disclaimer removal.

A DISTRIBUTION_MODE flag would re-enable disclaimers, soften verdicts to comparisons, and gate release behind a legal review.

- If distribution with named-security advice is ever wanted, that is an RIA registration/partnership decision (fee-capped, exam and disclosure obligations) — recorded here as a future fork, not a v1 concern.